ss-RF Simulation Package

Spin-1 Population Physics, Spectral Spin Diffusion, Tensor-Polarization Design, and Implementation of the Production Release

Technical documentation for the UVA simulation software

Abstract

This note documents the physical model, its derivation, its numerical implementation, and the tools built on the selective semi-saturation radio-frequency (ss-RF) simulation package for spin-1 continuous-wave NMR, in its production release (`ssRF`). The modelled observable is the overlapping Pake doublet of a quadrupolar spin-1 ensemble, with deuterons in solid polarized targets (d-butanol, ND_3) as the motivating case. The state is resolved into the three Zeeman-level populations of a set of orientation/frequency packets and, optionally, of two chemical sites (C–D and O–D) together with the inverse temperature of the nuclear dipolar reservoir.

The model contains:

- an analytic broadened Pake lineshape and a fitted two-site butanol lineshape;
- signed vector and tensor polarization with exact direct-hole and mirror-peak bookkeeping;
- an RF operator with a finite homogeneous linewidth, driven by arbitrary per-bin, time-scheduled programs, with separately tunable per-spin coupling and width profiles;
- spectral spin diffusion as mass-action pair currents for the three spin-1 flip-flop channels (same-branch, cross-branch, and an effective double-quantum channel) on a nonlocal Lorentzian zero-quantum kernel;
- DNP and spin-lattice relaxation;
- instantaneous adiabatic fast passage;
- a first-order Provotorov coupling to the dipolar reservoir.

Conservation of spins, of vector polarization and (where appropriate) of energy follows from the structure of the currents rather than from projections. The unique stationary state of the internal dynamics, a Boltzmann line at the remaining polarization, is derived rather than imposed.

The note then documents:

- the time integrator;
- the tensor-polarization optimizer, a discrete-adjoint, constrained search for the RF program that maximizes Q under power and energy limits;
- the calibrated d-butanol materials;
- a command-line training-data generator, including a closed-loop mode that reproduces the operating cycle;
- the machine-learning models trained on its output;
- the graphical interface, the regression tests, and a staged calibration procedure.

It also quantifies the peak–pedestal saturation balance with the calibrated materials. It supersedes the first-release note, *Real-Time ss-RF Simulation Package: Spin-Capacity-Weighted DNP, Depolarization, and Recovery*, and section 24 lists what changed and why. Exact spin-1 bookkeeping is kept separate from phenomenological closures throughout.

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1 Scope, model status, and central physical picture

The ss-RF simulation package is a forward model for the evolution of a spin-1 Pake-doublet NMR signal under selective RF, adiabatic fast passage (AFP), DNP, spectral spin diffusion, and spin-lattice relaxation. It also contains an inverse model: a designer that computes the RF program which maximizes the tensor polarization from a given state. It is not a first-principles many-body calculation. It is a physically constrained, experimentally parameterizable simulator, calibrated for deuterated butanol, intended to answer questions such as:

- What happens to the two overlapping absorption components when RF of finite linewidth is applied at one or many physical frequencies?
- How large is the direct hole in each branch, where do the corresponding mirror peaks appear, and how does a finite RF linewidth modify the classical one-half mirror rule?
- How does a hole recover with the microwaves off, when vector polarization removed by ss-RF cannot be recreated by spin diffusion, and how does recovery depend on which flip-flop channels operate?
- How does the answer change when DNP is on and supplies polarization from the electron-spin reservoir?
- Which RF program, within given power and energy limits, maximizes the tensor polarization from the present state, and how does that program change as the state evolves?
- Why does a hole near a Pake peak need a different RF power than one in the pedestal, and which material parameters decide this?
- Which rates must be measured, as functions of frequency, polarization, RF power, temperature and material, to make the predictions quantitative?

The physical basis follows the spin-1 population and Pake-doublet formalism developed for RF-manipulated solid targets in Refs. [21–24], together with standard solid-state NMR, spin-temperature and DNP theory [3–5, 9–11]. The characteristic powder-pattern density originates in the orientation-to-frequency mapping first associated with Pake-type spectra [1].

The spin-diffusion architecture follows the orientation-resolved model introduced in *Selective semi-saturation measurement of spin-1* [24]. That work formulates the level populations as $n_i(\theta, t)$, lets neighbouring orientation packets exchange polarization, and proposes a T_2-limited spectral-overlap kernel. The first release reduced this to a local relaxation plus a nearest-neighbour flux. The production release implements the spectral-overlap idea directly. Every pair of packets exchanges population through the microscopic spin-1 flip-flop channels, at a rate set by their zero-quantum spectral overlap and by the instantaneous availability of partners in the right levels [15–17].

The package rests on six central statements.

1. A spin-1 packet has three populations (n_+, n_0, n_-). The two allowed single-quantum absorption intensities are population differences, $I^+ \propto n_+ - n_0$ and $I^- \propto n_0 - n_-$.
2. At one physical frequency R in the overlap region, $I^+(R)$ and $I^-(R)$ generally come from two different orientation packets. A common RF tone therefore drives two distinct transitions with one common applied power.
3. Each orientation packet is observed twice: once in one absorption branch at R, and once in the other branch at $-R$. The mirror peak is therefore not an independent reservoir; it is another observable projection of the same three populations.
4. ss-RF equalizes the addressed pair of levels. It depolarizes the usual monotone spin-temperature populations; for a general non-Boltzmann state its effect on P and Q follows from the population currents. Internal spin diffusion redistributes polarization but cannot create vector polarization. DNP and spin-lattice relaxation are the external sources and sinks.

5. An equal-width frequency bin does not contain the same number of spins at every R. This *capacity* μ_k, set by the Pake density, enters every current through the populations. It is distinct from the *rate per spin*. In the production release any additional dependence of a per-spin rate on frequency, orientation or site is an explicit, separately adjustable mechanism profile. It is never a hidden second use of the density.
6. Recovery is a set of pair processes with definite matrix elements and definite conservation laws. Same-branch flip-flops conserve both P and Q; cross-branch flip-flops conserve P but change Q; double-quantum exchange moves vector polarization between packets without frequency matching. Which channels dominate is the largest open material question for tensor enhancement.

Scientific status of the simulation closure. The following are structurally fixed by spin-1 quantum mechanics:

- the packet capacity μ_k, the population bookkeeping and the mirror relation;
- the definitions of P and Q;
- the equality of the two single-quantum RF matrix elements;
- the channel structure and conservation laws of the flip-flop currents;
- the uniqueness of the Boltzmann stationary state.

The following are phenomenological closures with calibrated or adjustable parameters:

- the zero-quantum overlap shape and width, and the effective double-quantum ratio;
- the per-spin RF selectivity;
- the DNP target law;
- the first-order dipolar-reservoir coupling.

For d-butanol the lineshape, DNP build-up, RF strength and exchange were fitted to UVA target data. Per-spin RF selectivity and the cross-branch ratio are not constrained by present data (section 19).

2 Spin-1 energy levels and the quadrupolar orientation dependence

2.1 High-field Hamiltonian

A deuteron has spin $I = 1$, so in a strong holding field it has three Zeeman sublevels labelled $m = +1, 0, -1$. In a non-cubic solid, the nuclear quadrupole moment couples to the local electric-field-gradient (EFG) tensor. To first order in the quadrupolar coupling,

$$\hat{H} = -\hbar\omega_d \hat{I}_z + \hbar\omega_q \,\Lambda(\theta,\phi)\left(3\hat{I}_z^2 - I(I+1)\right), \tag{1}$$

where ω_d is the deuteron Larmor angular frequency, ω_q sets the quadrupolar scale, and

$$\Lambda(\theta,\phi) = 3\cos^2\theta - 1 + \eta\sin^2\theta\cos 2\phi. \tag{2}$$

The angle θ is measured between the EFG principal axis and the holding field, and η is the EFG asymmetry [3, 5, 6]. For $\hat{H}_Q \propto e^2qQ$, $\omega_q = e^2qQ/(8\hbar)$ in the normalization of equation (1). Equivalent normalizations occur in the literature; the simulation is built around the dimensionless frequency coordinate below.

For $I = 1$, $3\hat{I}_z^2 - I(I+1)$ takes the value $+1$ for $m = \pm 1$ and -2 for $m = 0$. The level energies are therefore

$$E_{\pm 1} = \mp\hbar\omega_d + \hbar\omega_q\Lambda, \qquad E_0 = -2\hbar\omega_q\Lambda, \tag{3}$$

and the two allowed magnetic-dipole transitions,

$$m = +1 \leftrightarrow 0, \qquad m = 0 \leftrightarrow -1, \tag{4}$$

occur at $\omega_d \mp 3\omega_q\Lambda$. The $\Delta m = 2$ transition $+1 \leftrightarrow -1$ lies at $2\omega_d$ with *no* first-order quadrupolar shift. This fact underlies the double-quantum exchange channel of section 7.3. Because the quadrupolar contribution depends on θ, the transition frequencies vary across a polycrystalline or amorphous sample. Powder averaging therefore produces two broad, overlapping absorption components rather than two sharp lines [23, 24].

Rotating-frame level energies used by the code. Remove the Larmor term in the rotating frame, subtract a constant, and measure energy in units of $3\hbar\omega_q$. The level energies of a packet whose I^+ transition lies at R (equation (9)) are then

$$E_+ = E_- = -R, \qquad E_0 = 0. \tag{5}$$

The I^+ transition is at $E_0 - E_+ = R$, the I^- transition at $E_- - E_0 = -R$, and the double-quantum transition at $E_- - E_+ = 0$. The total offset (non-Larmor) energy of the spin system is

$$E_{\text{spin}} = -\sum_k R_k \, (n_{k+} + n_{k-}), \tag{6}$$

which the extended engine balances against the dipolar reservoir (section 9).

Magnetic-dipole matrix elements. In the high-field limit a transverse RF field couples through $\hat{I}_x$, with

$$|\langle 0|\hat{I}_x|+1\rangle|^2 = |\langle -1|\hat{I}_x|0\rangle|^2 = \frac{1}{2}, \qquad |\langle 0|\hat{I}_+|-1\rangle|^2 = |\langle +1|\hat{I}_+|0\rangle|^2 = 2. \tag{7}$$

Both single-quantum transitions therefore have the same coupling, and the EFG angle introduces no leading-order suppression of the magnetic-dipole drive at the powder peak [3].

2.2 Dimensionless frequency coordinate and software branch convention

The frequency coordinate is

$$R = \frac{\omega - \omega_d}{3\omega_q}. \tag{8}$$

The simulation software adopts the branch mapping of Ref. [23]:

$$R_+(\theta) = 1 - 3\cos^2\theta, \qquad -2 \le R_+ \le 1, \tag{9}$$

$$R_-(\theta) = 3\cos^2\theta - 1 = -R_+(\theta), \qquad -1 \le R_- \le 2, \tag{10}$$

for $\eta = 0$; with asymmetry, $R_+ = 1 - 3\cos^2\theta - \eta\sin^2\theta\cos 2\phi$. The labels + and − identify the two absorption branches as used by the code. The main consequences are independent of sign convention:

- $\theta = 90°$ maps to the I^+ horn at $R = +1$ and to the I^- horn at $R = -1$.
- $\theta = 0$ maps to $R = -2$ for I^+ and $R = +2$ for I^-.
- The "magic" orientation $\cos^2\theta = 1/3$ maps to $R = 0$ in both branches.
- One orientation packet has two mirror-related spectral appearances, at R and $-R$.

In physical units, $R = 1$ is $3\omega_q/2\pi = \frac{3}{8}e^2qQ/h$. That is ≈ 63 kHz for C–D bonds with $e^2qQ/h \approx 167$ kHz. The O–D site of butanol has a larger coupling, so its line extends to $|R| \approx 2.7$ on the C–D scale. The broadened lineshape has tails beyond the strict first-order domains, so the grid extends beyond ±2: ±3 for the single-site demonstration material and ±6.19 for the d-butanol materials, which must contain the O–D component and its tails.

2.3 Why a Pake horn contains more spin capacity

For an isotropic powder, the number of EFG axes in a polar-angle interval is proportional to the solid-angle measure

$$\mathrm{d}N \propto \sin\theta\,\mathrm{d}\theta. \tag{11}$$

Equivalently, $u = \cos\theta$ is uniformly distributed on the hemisphere. For the I^+ branch,

$$R_+ = 1 - 3u^2, \qquad u = \sqrt{\frac{1-R_+}{3}}. \tag{12}$$

Therefore the ideal, unbroadened density per unit R is

$$\frac{\mathrm{d}N}{\mathrm{d}R_+} \propto \left|\frac{\mathrm{d}u}{\mathrm{d}R_+}\right| = \frac{1}{2\sqrt{3}}\frac{1}{\sqrt{1-R_+}}, \qquad \int_{-2}^{1}\frac{\mathrm{d}R_+}{2\sqrt{3}\sqrt{1-R_+}} = 1. \tag{13}$$

This Jacobian diverges at $R_+ = 1$, corresponding to $\theta = 90°$. Broadening makes the horn finite, but the physical point remains: many orientations are compressed into a small frequency interval. The I^- branch is the mirror image and has its horn at $R = -1$. This powder-density enhancement is the geometric origin of the Pake horns [1, 3].

The Jacobian counts *spins per unit frequency*. It does not set a *rate per spin*. The golden-rule rate of section 6.2 contains no factor of the density, and a flip-flop current between two packets contains the capacities of both packets exactly once, through their populations. Any further dependence of a per-spin rate on frequency must come from a separate physical mechanism, for example an orientation-dependent T_2. The production release models such mechanisms with the explicit profiles of section 6.4.

3 Populations, vector and tensor polarization, and NMR observables

3.1 Three-level populations

Let the normalized total populations of the three Zeeman levels be

$$n_+ + n_0 + n_- = 1. \tag{14}$$

The vector and tensor polarizations are

$$P = n_+ - n_-, \tag{15}$$

$$Q = n_+ - 2n_0 + n_- = 1 - 3n_0. \tag{16}$$

Solving for the level fractions gives

$$p_+(P,Q) = \frac{1}{3} + \frac{P}{2} + \frac{Q}{6}, \tag{17}$$

$$p_0(P,Q) = \frac{1-Q}{3}, \tag{18}$$

$$p_-(P,Q) = \frac{1}{3} - \frac{P}{2} + \frac{Q}{6}. \tag{19}$$

The physical region is the triangle in (P,Q) for which all three fractions are nonnegative: $Q \le 1$, $Q \ge 3|P| - 2$. The backend accepts an optional initial Q_0; materials and the GUI initialize with the Boltzmann value unless a non-Boltzmann state is loaded from a snapshot.

3.2 Boltzmann or single-spin-temperature relation

For a spin-1 system with one Zeeman spin temperature, $p_m \propto e^{\beta m}$. Writing $r = e^{\beta}$,

$$P = \frac{r - r^{-1}}{r + 1 + r^{-1}}, \qquad Q = \frac{r + r^{-1} - 2}{r + 1 + r^{-1}}. \tag{20}$$

Eliminating r (with $x = r + r^{-1}$, so $Q = (x-2)/(x+1)$ and $P^2 = (x^2-4)/(x+1)^2$) gives

$$Q_{\mathrm{B}}(P) = 2 - \sqrt{4 - 3P^2}. \tag{21}$$

For example, $Q_{\mathrm{B}}(0.50) = 0.1972$. This relation is used to construct the default initial state, the DNP and T_1 targets, and the targets in the data generator. It is *not* imposed on the evolving state, and it is no longer used as a recovery reference. In the production release, the approach of an isolated state to the Boltzmann line follows from the flip-flop currents (section 7.4). During RF manipulation $Q(t)$ can differ from $Q_{\mathrm{B}}[P(t)]$, and the GUI displays both.

The branch population differences in a uniform Boltzmann state are

$$p_+ - p_0 = \frac{P + Q}{2}, \tag{22}$$

$$p_0 - p_- = \frac{P - Q}{2}. \tag{23}$$

Their sum is P and their difference is Q. The corresponding equilibrium branch-area ratio is

$$r(P) = \frac{I^+}{I^-} = \frac{\sqrt{4 - 3P^2} + P}{2 - 2P}, \tag{24}$$

which is the ratio used in the 2023 formalism [23].

3.3 Absorption intensities

For an orientation packet at θ, the absorptive components are

$$I^+(\theta) = C\,[\rho_+(\theta) - \rho_0(\theta)], \tag{25}$$

$$I^-(\theta) = C\,[\rho_0(\theta) - \rho_-(\theta)], \tag{26}$$

where C is the NMR calibration factor and the ρ_m are population densities. Integrated over the full branches,

$$P = C\,(I^+ + I^-), \tag{27}$$

$$Q = C\,(I^+ - I^-). \tag{28}$$

These area relations are definitions and hold for any state. What fails out of equilibrium is the inference of Q from P using equation (21) [23, 24]. That failure is why a learned polarization model is useful for manipulated states (section 21).

Signed polarization is handled naturally. When $P < 0$, the ordering of the three populations reverses and both absorptive population differences can become negative. The software plots signed branch intensities, and every rate equation is written for signed populations.

4 Packet space, physical-frequency overlap, and mirror bookkeeping

4.1 State coordinate used by the simulation software

The dynamic state is stored on a one-dimensional packet coordinate

$$x_k \equiv R_+(\theta_k), \qquad k = 0, \ldots, N-1, \tag{29}$$

on a uniform grid of width ΔR. Each packet contains a three-component population vector

$$\boldsymbol{n}_k = (n_{k+}, n_{k0}, n_{k-})^{\mathsf{T}}, \qquad n_{k+} + n_{k0} + n_{k-} = \mu_k, \qquad \sum_k \mu_k = 1. \tag{30}$$

The packet is observed as

$$I^+ \text{ at physical } R = x_k, \qquad I^- \text{ at physical } R = -x_k. \tag{31}$$

Thus the two branch appearances are not separate states.

Symmetric grid and the mirror index. The production release requires the grid to be symmetric about zero, to 10^{-12}. Then the packet whose I^+ line is at $-x_k$ is

$$\bar{k} \equiv N - 1 - k, \qquad x_{\bar{k}} = -x_k, \tag{32}$$

and every mirror lookup is an exact index reversal, with no interpolation. `IdealBinModel.reset` enforces the symmetry. The d-butanol materials use $N = 500$ on $[-6.186, 6.186]$ ($\Delta R = 0.0248$); the demonstration material uses $N = 500$ on $[-3, 3]$.

4.2 Two different packets at one physical RF frequency

At a physical frequency R_b the two overlapping components are

$$I^+(R_b) : \text{packet } k_+ \text{ with } x_{k_+} \simeq R_b, \tag{33}$$

$$I^-(R_b) : \text{packet } k_- = \overline{k_+} \text{ with } x_{k_-} \simeq -R_b. \tag{34}$$

These are generally two different values of θ. A single RF tone has one amplitude, but it acts on both transitions because both are resonant at R_b. With the matrix elements of equation (7), the two transitions have the same golden-rule rate per spin (section 6.2). Their currents differ because their population differences differ, and because of any explicit per-spin profile.

The mirror observables are

$$I^-(-R_b) \text{ from packet } k_+, \tag{35}$$

$$I^+(-R_b) \text{ from packet } k_-. \tag{36}$$

This bookkeeping is essential: direct holes and mirror peaks must be calculated from shared populations, not evolved as four independent signals.

5 Static lineshape and packet capacity

5.1 Analytic broadened single-site branch function

The single-site lineshape (`line_model = single_pake`) is the analytic broadened Pake function of Refs. [20, 23]. Define

$$s \equiv \eta \cos 2\phi, \tag{37}$$

$$X_2(R,\epsilon) \equiv \sqrt{\Gamma_{\mathrm{line}}^2 + (1 - \epsilon R - s)^2}, \tag{38}$$

$$X(R,\epsilon) \equiv \sqrt{X_2(R,\epsilon)}, \tag{39}$$

$$Y \equiv \sqrt{3 - s}, \tag{40}$$

$$\cos\alpha \equiv \frac{1 - \epsilon R - s}{X_2(R,\epsilon)}. \tag{41}$$

Then the branch density is

$$f(R,\epsilon) = \frac{1}{2\pi X}\left\{2\cos\frac{\alpha}{2}\left[\tan^{-1}\left(\frac{Y^2 - X_2}{2YX\sin(\alpha/2)}\right) + \frac{\pi}{2}\right] + \sin\frac{\alpha}{2}\ln\left[\frac{Y^2 + X_2 + 2YX\cos(\alpha/2)}{Y^2 + X_2 - 2YX\cos(\alpha/2)}\right]\right\}. \tag{42}$$

The branch $\epsilon = +1$ has its sharp horn on the positive-R side, and $f(R,-1) = f(-R,+1)$. The demonstration material uses $\Gamma_{\mathrm{line}} = 0.05$ and $s = 0.04$, approximate ND_3 values [23].

5.2 Azimuth-averaged resolvent form for general asymmetry

For the fitted materials (`line_model = multisite_butanol`), each site uses the full azimuthal average. Write $c = \cos\theta$ and $y_\phi^2 = 3 - \eta\cos 2\phi$. Then the static offset is $x = 1 - \eta\cos 2\phi - y_\phi^2 c^2$. A Lorentzian broadening of half-width γ is the imaginary part of the resolvent:

$$g_\gamma(x) = -\frac{1}{\pi}\,\mathrm{Im}\left\langle \int_0^1 \frac{\mathrm{d}c}{1 - \epsilon x - \eta\cos 2\phi - y_\phi^2 c^2 + i\gamma}\right\rangle_\phi = -\frac{1}{\pi}\,\mathrm{Im}\left\langle \frac{\operatorname{artanh}(y_\phi/r_\phi)}{y_\phi\, r_\phi}\right\rangle_\phi, \tag{43}$$

with $r_\phi = \sqrt{1 - \epsilon x - \eta\cos 2\phi + i\gamma}$. The c-integral is elementary ($\int_0^1 \mathrm{d}c/(r^2 - y^2c^2) = \operatorname{artanh}(y/r)/(yr)$). Only the average over $\phi \in [0, \pi/2]$ is done numerically, by Gauss–Legendre quadrature (`line_phi_points`; code default 64, 128 in the d-butanol materials). For $\epsilon = -1$ the code conjugates the kernel, which mirrors the branch. The form is exact for a Lorentzian convolution of the first-order powder pattern, and has unit integrated absorption on the infinite axis.

5.3 Two-site butanol lineshape

Butanol-d_{10} contains C–D and O–D deuterons with different quadrupole couplings. Let a_{CD} be the C–D scale (one unit of R), $a_{\mathrm{OD}} = \rho_{\mathrm{OD}} a_{\mathrm{CD}}$ and f_{OD} the O–D fraction. The capacity density is

$$g(R) = \frac{1 - f_{\mathrm{OD}}}{a_{\mathrm{CD}}}\, g_{\gamma/a_{\mathrm{CD}}}^{(\eta_{\mathrm{CD}})}\!\left(\frac{R}{a_{\mathrm{CD}}}\right) + \frac{f_{\mathrm{OD}}}{a_{\mathrm{OD}}}\, g_{\gamma/a_{\mathrm{OD}}}^{(\eta_{\mathrm{OD}})}\!\left(\frac{R}{a_{\mathrm{OD}}}\right), \tag{44}$$

so that f_{OD} is an integrated site weight and both sites share one physical width γ. The fit to the 2014 UVA spectra (held-out RMS 0.79%) gives $\rho_{\mathrm{OD}} = 1.355$, $f_{\mathrm{OD}} = 0.021$, $\eta_{\mathrm{CD}} = 0.065$, $\eta_{\mathrm{OD}} = 0.033$ and $\gamma = 0.0236$.

Mechanisms that depend on orientation need each packet's angle and site composition. The code inverts the $\eta = 0$ map per site, $\cos^2\theta_s(R) = \mathrm{clip}((1 - R/a_s)/3, 0, 1)$, and records the O–D share $w_{\mathrm{OD}}(R) = g_{\mathrm{OD}}/(g_{\mathrm{CD}} + g_{\mathrm{OD}})$ (`packet_site_geometry`).

5.4 Normalization and packet capacity

The $\epsilon = +1$ density is normalized on the simulation grid and converted to packet fractions,

$$\rho_k = \frac{g(x_k)}{\int g(x)\,\mathrm{d}x}, \qquad \mu_k = \frac{\rho_k \Delta R}{\sum_j \rho_j \Delta R}. \tag{45}$$

The initial packet populations are

$$n_{km}(0) = \mu_k p_m(P_0, Q_0). \tag{46}$$

Thus μ_k already changes the absolute number of spins and the absolute signal in each bin. A bin with larger μ_k contains more population to be moved by every process. The displayed intensities are

$$I^+(x_k) = \mathcal{C}_{\mathrm{disp}} \frac{n_{k+} - n_{k0}}{\Delta R}, \tag{47}$$

$$I^-(-x_k) = \mathcal{C}_{\mathrm{disp}} \frac{n_{k0} - n_{k-}}{\Delta R}. \tag{48}$$

The display calibration $\mathcal{C}_{\mathrm{disp}}$ affects only plotted units.

5.5 Capacity is not identical to a microscopic rate

A central interpretive distinction is

$$\underbrace{\text{observable current}}_{\mathrm{d}I/\mathrm{d}t} \sim \underbrace{\text{number of spins}}_{\mu_k} \times \underbrace{\text{population imbalance per spin}}_{p_a - p_b} \times \underbrace{\text{transition rate per spin}}_{W_k}. \tag{49}$$

Because $n_{ka} = \mu_k p_{ka}$, the capacity is already present in the state. With a uniform per-spin rate, a horn bin still produces a larger *absolute* source or sink, because it contains more spins. Its *fractional* saturation proceeds at the same rate as anywhere else.

The first release multiplied every rate by an extra weight $w_k \propto \rho_k^\alpha$. With $\alpha = 1$ this made absolute horn currents scale roughly as ρ^2. The production release removes that double use:

- RF and spin-diffusion currents contain the capacity once.
- Any further frequency, orientation or site dependence of a per-spin rate is an explicit mechanism profile with its own parameters (section 6.4).
- The old weight survives only as the DNP build-rate weighting (`capacity_rate_power`, section 8) and in the legacy RF modes kept for comparison.

Practical interpretation. A fractional quantity (hole depth relative to the local line, fractional build-up, required RF power for a given fractional saturation) is independent of μ_k unless a per-spin rate depends on k. A difference between peak and pedestal in such a quantity therefore measures a per-spin difference in RF response, in replenishment, or in both. Section 18.7 shows how to separate them, and what the calibrated materials predict.

6 Complete rate model implemented in the simulation software

6.1 Additive mechanism decomposition

Without the extended engine (section 9), the model evaluates the population derivative as

$$\dot{\boldsymbol{n}} = \dot{\boldsymbol{n}}_{\mathrm{RF}} + \dot{\boldsymbol{n}}_{\mathrm{SQ}} + \dot{\boldsymbol{n}}_{\mathrm{X}} + \dot{\boldsymbol{n}}_{\mathrm{DQ}} + \dot{\boldsymbol{n}}_{\mathrm{DNP}} + \dot{\boldsymbol{n}}_{T_1}, \tag{50}$$

with AFP applied as a discrete map between integration steps (section 10). The terms have distinct physical roles:

Term	Physical role	Conserves
ss-RF	Incoherent saturation of every transition within the homogeneous linewidth of each carrier	μ_k; not P, not Q
Same-branch (SQ)	Flip-flops $\|+0\rangle \leftrightarrow \|0+\rangle$ and $\|0-\rangle \leftrightarrow \|-0\rangle$ between packets with overlapping transitions	μ_k, $N^{\pm,0}$, P, Q
Cross-branch (X)	Flip-flops $\|00\rangle \leftrightarrow \|+-\rangle$ between packets whose I^+ and I^- lines overlap	μ_k, P; changes Q
Double quantum (DQ)	Effective $\|+-\rangle \leftrightarrow \|-+\rangle$ exchange with no first-order frequency condition	μ_k, $N^{\pm,0}$, P, Q
DNP	Relaxation of every packet toward the Boltzmann fractions at P_{sat}	μ_k
T_1	Relaxation toward the Boltzmann fractions at P_{eq}	μ_k

Here $N^m = \sum_k n_{km}$. No term resets the state toward a reference line, and no term is projected after the fact. Every conservation property in the table follows from the form of the currents.

6.2 RF saturation: the golden-rule rate

A linearly polarized field $2B_1 \cos \omega t$ has a co-rotating component $\hat{H}_1 = \frac{\hbar\omega_1}{2}(\hat{I}_+ e^{-i\omega t} + \text{h.c.})$ with $\omega_1 = \gamma_d B_1$. Suppose the coherence between the levels a and b of a transition decays exponentially at rate Γ (the homogeneous half-width, a T_2), and the drive is weak, $\omega_1 \ll \Gamma$. The coherence then follows the populations adiabatically, and eliminating it gives a rate equation $\dot{n}_a = -W(n_a - n_b)$ [2–4, 7] with

$$W = 2\pi \frac{\omega_1^2}{4} |\langle b|\hat{I}_+|a\rangle|^2 \, g_{\mathrm{L}}(\Delta) = \omega_1^2 \frac{\Gamma}{\Gamma^2 + \Delta^2}, \tag{51}$$

where g_{L} is the unit-area Lorentzian and the last step uses equation (7). Both spin-1 transitions have this same rate, twice the spin-$\frac{1}{2}$ value. Equivalently, the population difference decays at $\lambda_{\mathrm{RF}} = 2W$, which is the Bloch-rate form $\Omega_1^2 T_2/(1 + \Delta\omega^2 T_2^2)$ with $\Omega_1^2 = 2\omega_1^2$.

Validity. In the code, carrier j has power $U_j \propto B_1^2$, and the material constant κ (`rf_spectral_strength`) absorbs the conversion $\omega_1^2 \leftrightarrow U$ and the R unit:

$$W_{kj} = \kappa \frac{\Gamma}{\Gamma^2 + (x_k - x_j)^2} U_j. \tag{52}$$

For d-butanol, $\kappa = 0.0196$ and $\Gamma = 0.015$ give an on-resonance rate $\kappa/\Gamma = 1.31\,\mathrm{s}^{-1}$ per unit U. With $\Gamma \approx 0.94\,\mathrm{kHz}$ in physical units, this corresponds to $\omega_1/2\pi \approx 14\,\mathrm{Hz}\sqrt{U}$. That is orders of magnitude below Γ, so the weak-drive closure is self-consistent for every command the designer produces. Coherent phenomena (nutation, spin locking, rotary saturation beyond the Provotorov terms) are absent.

6.3 The discrete RF operator

Packet k sees the RF at its I^+ frequency x_k and at its I^- frequency $-x_k$:

$$W_k^+ = c_k \sum_j K_k(x_k - x_j)\, U_j, \qquad W_k^- = c_k \sum_j K_k(-x_k - x_j)\, U_j, \qquad K_k(\delta) = \frac{\kappa\, \Gamma_k}{\Gamma_k^2 + \delta^2}. \tag{53}$$

c_k is the per-packet coupling and Γ_k the per-packet homogeneous width (section 6.4); by default c_k is fixed by the peak selectivity and $\Gamma_k = \Gamma$. The RF currents are

$$J_k^+ = W_k^+(n_{k+} - n_{k0}), \quad J_k^- = W_k^-(n_{k0} - n_{k-}), \tag{54}$$

$$\dot{n}_{k+}\big|_{\mathrm{RF}} = -J_k^+, \qquad \dot{n}_{k0}\big|_{\mathrm{RF}} = J_k^+ - J_k^-, \qquad \dot{n}_{k-}\big|_{\mathrm{RF}} = J_k^-. \tag{55}$$

There is no row normalization, no edge correction and no ΔR factor: U_j is a carrier power, not a density. Refining the grid therefore requires the *total* power of a distributed command to be preserved. In the limit $\Gamma \to 0$, with one carrier exactly on bin b, equation (53) reduces to the ideal-bin operator of the first release: $W_k^+ \propto \delta_{kb}$ and $W_k^- \propto \delta_{k\bar{b}}$.

Implementation. With uniform Γ, $K(x_k - x_j)$ is a symmetric Toeplitz matrix. The product KU is evaluated by FFT on a zero-padded circulant embedding of length $L \geq 2N-1$, which avoids wrap-around. On the symmetric grid, $\sum_j K(-x_k - x_j)U_j = (KU)_{\bar{k}}$, so W^- is the reversed W^+ field and costs nothing extra. The transpose needed by the optimizer's adjoint is the same convolution. The last command's response is cached, because one command is held for many steps. With per-packet widths, the kernels $K^{\pm}$ are built as dense $N \times N$ matrices (`RFRateOperator`).

6.4 Per-spin coupling, selectivity, and mechanism profiles

Equation (51) contains no density of spins. Whether horn and pedestal spins have different per-spin RF rates (different T_2 or coupling) is a material question that present burn data do not settle. The model therefore provides two positive per-packet factors. Both multiply the *two* transitions of a packet, so each packet keeps its mirror relationship.

Peak selectivity and pedestal gain.

$$c_k = \Big[1 + (1 - \sigma_k)\big((\rho_k/\rho_{\max})^s - 1\big)\Big]\,[1 + (G_{\mathrm{p}} - 1)\sigma_k]\; f_k^{\mathrm{rf}}. \tag{56}$$

- σ_k is a cubic smoothstep in $||x_k| - R_c|$: 0 inside the peak bands (half-width `rf_peak_half_width_R` around $\pm R_c$) and 1 beyond the taper `rf_transition_width_R`.
- s is `rf_peak_selectivity`. The densest packet and the outer pedestal keep unit coupling.
- G_{p} is `rf_coupling_pedestal_gain`.

General mechanism profiles. Every mechanism $X \in \{\text{RF coupling, RF width, exchange, DNP, } T_1\}$ can carry the factor

$$f_k^X = D_k^X \sum_{s \in \{\mathrm{CD},\mathrm{OD}\}} w_s(x_k)\, m_s^X \left[1 + a_2^X P_2(\cos\theta_s) + a_4^X P_4(\cos\theta_s)\right], \tag{57}$$

where:

- $m_{\mathrm{CD}}^X = 1$ and m_{OD}^X is `X_od_factor`;
- $D_k^X = \mathrm{clip}\big((\rho_k/\bar{\rho}_\mu)^{e_X}, c^{-1}, c\big)$, renormalized to μ-weighted mean one;
- P_2 and P_4 are Legendre polynomials.

The angular form is the natural one for a coupling or a T_2 that depends on orientation, for example through the dipolar local field or a motional spectral density at the EFG angle. All defaults give $f^X \equiv 1$. The RF width enters as $\Gamma_k = \Gamma f_k^{\mathrm{width}}$ on the *receiving* packet. At fixed κ, a narrower packet has a higher on-resonance rate κ/Γ_k and a narrower hole. This is the physically motivated route to horn-favouring selectivity. The `d_butanol_basic` and `d_butanol_extend` materials use it (section 19).

These profiles implement the separately calibrated RF, recovery and DNP weights w_k^{RF}, w_k^{rec}, w_k^{DNP} that the first-release note proposed as a future extension.

Legacy RF modes. `rf_response_mode = legacy` reproduces the first-release law, in which RF is multiplied by the DNP density weight. `independent` adds a separate density exponent and a smooth pedestal gain; it is the MaterialTuning-V1 law. Both use one-bin rates, $W_k^+ = w_k U_k$ and $W_k^- = w_k U_{\bar{k}}$. They are kept to reproduce earlier results; the shipped materials use `spectral`.

6.5 How RF changes P and Q

From equations (15), (16) and (55),

$$\dot{P}|_{\mathrm{RF}} = -\sum_k (J_k^+ + J_k^-), \qquad \dot{Q}|_{\mathrm{RF}} = 3\sum_k (J_k^- - J_k^+) = 3\sum_k (W_k^- \Delta_k^- - W_k^+ \Delta_k^+), \tag{58}$$

with $\Delta_k^+ = n_{k+} - n_{k0}$ and $\Delta_k^- = n_{k0} - n_{k-}$. For the usual monotone spin-temperature populations, RF reduces $|P|$. It raises Q exactly when it saturates more I^- population difference than I^+ population difference. For one narrow carrier at R, the sign of $\dot{Q}$ is the sign of $c_{\overline{k_+}}\Delta_{\overline{k_+}}^- - c_{k_+}\Delta_{k_+}^+$. In words, the coupling-weighted I^- signal at that frequency must exceed the I^+ signal. At $P > 0$ this selects two regions:

- the little peak ($R \approx -0.93$, the I^- horn);
- the region $R \gtrsim 1.1$, where only the I^- line exists.

The optimizer's candidate mask (section 14.4) is this criterion evaluated with the full kernel.

6.6 Direct holes, mirror peaks, and the one-half gain rule

Using equations (47) and (48), take the ideal-bin limit: a carrier on bin b with $\Gamma \to 0$, so that only $J_{k_+}^+$ acts on packet k_+ and only $J_{k_-}^-$ on packet k_-. Write $\mathcal{K} = \mathcal{C}_{\mathrm{disp}}/\Delta R$. Then

$$\left.\frac{\mathrm{d}I^+(R_b)}{\mathrm{d}t}\right|_{\mathrm{RF}} = -2\mathcal{K}J_{k_+}^+, \qquad \left.\frac{\mathrm{d}I^-(-R_b)}{\mathrm{d}t}\right|_{\mathrm{RF}} = +\mathcal{K}J_{k_+}^+, \tag{59}$$

$$\left.\frac{\mathrm{d}I^-(R_b)}{\mathrm{d}t}\right|_{\mathrm{RF}} = -2\mathcal{K}J_{k_-}^-, \qquad \left.\frac{\mathrm{d}I^+(-R_b)}{\mathrm{d}t}\right|_{\mathrm{RF}} = +\mathcal{K}J_{k_-}^-, \tag{60}$$

and therefore

$$\dot{I^-}(-R_b) = -\frac{1}{2}\dot{I^+}(R_b), \qquad \dot{I^+}(-R_b) = -\frac{1}{2}\dot{I^-}(R_b). \tag{61}$$

The integrated mirror gain is one half of the direct loss [23]. The mirror peaks do not contradict depolarization: each direct transition loses twice the signal gained in the mirror projection, leaving a net reduction in branch area, $\dot{P}|_{\mathrm{RF}} = -J_{k_+}^+ - J_{k_-}^-$.

Finite linewidth. With $\Gamma > 0$, the two conditions behind the one-half rule break:

- a carrier near the line centre ($|R| \lesssim \Gamma$) drives *both* transitions of the same packet;
- the carrier's Lorentzian wings reach neighbouring packets.

The ratio is then no longer fixed, which a regression test checks. Spin diffusion, DNP and T_1 also act during and after a burn. A measured mirror/direct ratio that differs from $\frac{1}{2}$ is therefore a diagnostic of these processes, not a contradiction of the bookkeeping.

7 Spectral spin diffusion

7.1 Flip-flops between quadrupole-split spin-1 nuclei

The secular dipolar coupling between two like spins contains the flip-flop operator $\hat{I}_+^{(1)}\hat{I}_-^{(2)} + \mathrm{h.c.}$. For spin 1 it connects three kinds of pair states [3, 16, 17]. Using equation (5), the energy mismatch each

process must exchange with the dipolar bath, when packet i changes first, is:

$$\text{same-branch } (+): \quad |+_i 0_j\rangle \leftrightarrow |0_i +_j\rangle, \qquad \delta = x_i - x_j, \tag{62}$$

$$\text{same-branch } (-): \quad |0_i -_j\rangle \leftrightarrow |-_i 0_j\rangle, \qquad \delta = -(x_i - x_j), \tag{63}$$

$$\text{cross-branch}: \quad |0_i 0_j\rangle \leftrightarrow |+_i -_j\rangle, \qquad \delta = x_i + x_j. \tag{64}$$

The squared matrix elements are equal:

$$|\langle 0+|\hat{I}_+^{(2)}\hat{I}_-^{(1)}|+0\rangle|^2 = |\langle +-|\hat{I}_+^{(1)}\hat{I}_-^{(2)}|00\rangle|^2 = 4. \tag{65}$$

For a spatially random dipolar network, the cross-branch to same-branch ratio r_{x} (`cross_branch_ratio`) is therefore 1, which is the default. Same-branch exchange needs $x_i \approx x_j$; cross-branch exchange needs $x_i \approx -x_j$, so that the I^+ line of one packet overlaps the I^- line of the other.

A flip-flop with mismatch δ proceeds at a rate proportional to the zero-quantum (ZQ) spectral density at δ. That density is modelled as a normalized-to-one overlap,

$$S(\delta) = \frac{1}{1 + (\delta/w)^2} \quad \text{(Lorentzian, default)} \qquad \text{or} \qquad S(\delta) = e^{-\delta^2/2w^2}, \tag{66}$$

with width w given by `zq_width_R`. A Lorentzian follows from an exponentially decaying ZQ correlation function. Its algebraic tails allow weak exchange far off resonance, with the energy taken by the dipolar bath. Outside the extended engine, that bath is treated as infinitely hot and infinitely large. w is *not* the static line width γ, nor the RF homogeneous width Γ. A physical estimate is $w \approx 2\Gamma$, the convolution of two homogeneous lines.

7.2 Pair currents

Treat each packet as well mixed, and assume that the spatial neighbourhood of a spin does not depend on its EFG orientation (`orientation_corr_fraction` $= 0$). The forward rate of $(+_i, 0_j) \to (0_i, +_j)$ is then $K_0 \mu_i \mu_j S_{ij}\, p_{i+} p_{j0}$, where $S_{ij} = S(x_i - x_j)$ and K_0 is `diffusion_scale`. The net currents are of mass-action form:

$$J_{ij}^{(+)} = K_0 S_{ij}(n_{i0} n_{j+} - n_{i+} n_{j0}), \tag{67}$$

$$J_{ij}^{(-)} = K_0 S_{ij}(n_{i-} n_{j0} - n_{i0} n_{j-}), \tag{68}$$

$$J_{ij}^{(\mathrm{x})} = r_{\mathrm{x}} K_0 X_{ij}(n_{i0} n_{j0} - n_{i+} n_{j-}), \qquad X_{ij} = S(x_i + x_j)(1 - \tfrac{1}{2}\delta_{ij}). \tag{69}$$

The factor $\frac{1}{2}$ on the diagonal of X counts unordered pairs drawn from one packet; it matters only near $R = 0$. The packet updates are

$$\begin{aligned} \dot{n}_{i+}\big|_{\mathrm{SQ+X}} &= \textstyle\sum_j J_{ij}^{(+)} + \sum_j J_{ij}^{(\mathrm{x})}, \\ \dot{n}_{i-}\big|_{\mathrm{SQ+X}} &= -\textstyle\sum_j J_{ij}^{(-)} + \sum_j J_{ji}^{(\mathrm{x})}, \\ \dot{n}_{i0}\big|_{\mathrm{SQ+X}} &= -\dot{n}_{i+} - \dot{n}_{i-}. \end{aligned} \tag{70}$$

The capacities appear exactly once, through $n = \mu p$. The Pake density therefore sets the number of *partners* a packet has within the ZQ width, and is never applied again as a rate multiplier.

Microwave factor and per-packet exchange profiles. With DNP on, K_0 is multiplied by `microwave_diffusion_factor` (default 1), which allows for diffusion modified near paramagnetic centres. The exchange profile of equation (57) enters as $S_{ij} \to \sqrt{f_i^{\mathrm{ex}} f_j^{\mathrm{ex}}}\, S_{ij}$. This keeps every kernel symmetric and every current reversible.

Orientation correlation. If the real-space neighbourhood correlates with EFG orientation, the kernel is multiplied by $(1 - f_\theta) + f_\theta\, e^{-(\theta_i-\theta_j)^2/2\sigma_\theta^2}$. This is optional and off in all shipped materials.

7.3 Double-quantum transport

The process $(+_i, -_j) \leftrightarrow (-_i, +_j)$ exchanges vector polarization between packets. It changes neither $N^\pm$ nor E_{spin} (equation (5)), so it needs no single-quantum frequency matching. Microscopically it is a second-order process: two flip-flops through a virtual intermediate state, whose amplitude involves the mean squared coupling. It is folded into one effective ratio r_{DQ} (`double_quantum_ratio`) to K_0:

$$J_{ij}^{\text{DQ}} = r_{\text{DQ}} K_0 \sqrt{f_i^{\text{ex}} f_j^{\text{ex}}}\, C_{ij} (n_{i-} n_{j+} - n_{i+} n_{j-}), \qquad \dot{n}_{i+} \mathrel{+}= \textstyle\sum_j J_{ij}^{\text{DQ}}, \quad \dot{n}_{i-} \mathrel{-}= \textstyle\sum_j J_{ij}^{\text{DQ}}, \tag{71}$$

where C_{ij} is a spatial/orientation correlation matrix, unity off the diagonal by default. For $C \equiv 1$ the sum separates into $\sum_j n_{j\pm}$, so the cost is $O(N)$. In the fitted d-butanol material ($r_{\text{DQ}} = 3.03$, broad $w = 0.717$), DQ is a dominant refill channel. Refilling is then largely nonlocal in frequency (section 19).

7.4 Conservation laws, H-theorem, and the stationary state

Invariants.

- Each same-branch event conserves N^+, N^0 and N^-, and hence P and Q.
- A cross-branch event $(0, 0) \to (+, -)$ changes (N^+, N^0, N^-) by $(+1, -2, +1)$. It conserves P and changes Q by $+6$ per event, in units of the packet population.
- DQ events conserve $N^\pm$.
- All channels conserve every μ_k.

So the internal dynamics never changes P, and it changes Q only through the cross-branch channel.

H-theorem. Write each channel as $\sum_{\text{pairs}} \kappa\,(a - b)$ with $\kappa \geq 0$, where a and b are the forward and backward products of fractions. Then

$$\frac{\mathrm{d}}{\mathrm{d}t} \sum_{k,m} n_{km} \ln p_{km} = \sum_{\text{pairs}} \kappa\,(a - b) \ln \frac{a}{b} \;\geq\; 0, \tag{72}$$

so a relative entropy is a Lyapunov function, and stationary states satisfy detailed balance pair by pair:

- same-branch: p_{i+}/p_{i0} and p_{i-}/p_{i0} are equal for all packets connected by $S > 0$;
- cross-branch: $p_0^2 = p_+ p_-$;
- DQ: p_{i+}/p_{i-} is uniform.

On a connected graph with $r_{\text{x}} > 0$, these conditions force $p_{km} \propto r^m$ with one r for all packets, which is a Boltzmann state. Since P is conserved, the stationary state is unique: *the Boltzmann line at the remaining P*.

The first release imposed this behaviour through a reference state $\boldsymbol{n}^{\text{B}}[P(t)]$ and a projection. Here it is a theorem about the currents. The tests check it directly: stationarity of every channel at Boltzmann, positive entropy production, relaxation of arbitrary states to the unique Boltzmann line at the remaining P, and non-stationarity of uniform non-Boltzmann states.

Restricted channels. Without the cross channel ($r_{\text{x}} = 0$), every state with uniform fractions is stationary, including non-Boltzmann ones, and Q is frozen at fixed P. This is the `d_butanol_same_branch_transport` hypothesis. It is physically extreme, but it is the regime in which steady RF + DNP gives the largest tensor enhancement (section 11).

7.5 Fast evaluation: Toeplitz and Hankel products

On the symmetric uniform grid, $S(x_i - x_j)$ depends only on $i - j$, so it is Toeplitz. Because $-x_j = x_{\bar{\jmath}}$,

$$\sum_j S(x_i + x_j)\, v_j = \sum_j S(x_i - x_{\bar{\jmath}})\, v_j = \left(S\, v^{\mathrm{rev}}\right)_i, \tag{73}$$

so the cross kernel is the same Toeplitz product applied to the reversed vector, minus half the diagonal. Expanding the mass-action currents gives, for example,

$$\textstyle\sum_j J_{ij}^{(+)} = K_0\left[n_{i0}(Sn_+)_i - n_{i+}(Sn_0)_i\right], \tag{74}$$

and the $i = j$ terms cancel identically. All three population columns are convolved in one batched real FFT. `Spin1Model._spin_diffusion_terms` provides three algebraically identical evaluators, and the tests require agreement to rounding:

- FFT, the default with no orientation correlation and no ambiguous hard cutoff;
- sparse CSR matrices;
- explicit pair lists with `np.add.at`.

The default keeps the full Lorentzian tails on the finite grid; a positive `kernel_cutoff_widths` gives a banded kernel.

8 DNP and spin-lattice relaxation

8.1 DNP source term

DNP drives each packet toward the Boltzmann fractions at a chosen saturation polarization:

$$\dot{\boldsymbol{n}}_k\big|_{\mathrm{DNP}} = k_{\mathrm{D}}\, w_k^{\mathrm{D}} \left(\mu_k\, \boldsymbol{p}^{\mathrm{B}}(P_{\mathrm{sat}}) - \boldsymbol{n}_k\right), \qquad w_k^{\mathrm{D}} = \frac{\mathrm{clip}\left((\rho_k/\bar{\rho}_\mu)^{\alpha_{\mathrm{D}}}, c^{-1}, c\right)}{\sum_j \mu_j\, \mathrm{clip}\left((\rho_j/\bar{\rho}_\mu)^{\alpha_{\mathrm{D}}}, c^{-1}, c\right)}\, f_k^{\mathrm{dnp}}, \tag{75}$$

with $\alpha_{\mathrm{D}} =$ `capacity_rate_power` and $c =$ `capacity_rate_clip`. Because the target and the packet each sum to μ_k, DNP redistributes populations within a packet without changing its spin count. With RF and T_1 absent, every packet approaches the same target fractions, and the total polarization approaches P_{sat} without overshoot. The density weight makes horn packets approach the target faster during the transient. The global build-up is then multi-exponential, with

$$\dot{P}\big|_{\mathrm{DNP}} = k_{\mathrm{D}} \sum_k w_k^{\mathrm{D}} \{\mu_k [p_+^{\mathrm{B}} - p_-^{\mathrm{B}}](P_{\mathrm{sat}}) - (n_{k+} - n_{k-})\}. \tag{76}$$

Physical reading. In the thermal-mixing regime, the nuclear Zeeman system reaches a common spin temperature with the electron non-Zeeman reservoir [10, 12, 13]. The nuclear populations then approach a Boltzmann distribution that is the same at every orientation of a powder. Equation (75) is the simplest closure with that fixed point and one build-up rate. It does not separate the solid effect, the cross effect, thermal mixing or the differential solid effect, and it does not represent nuclear-spin diffusion barriers around paramagnetic centres. For d-butanol the fit to the 2014 build-up gives $k_{\mathrm{D}} = 2.30 \times 10^{-3}\,\mathrm{s}^{-1}$ ($\tau \approx 435\,\mathrm{s}$) and $P_{\mathrm{sat}} = 0.476$ (held-out RMS 0.79%).

8.2 Optional longitudinal relaxation

$$\dot{\boldsymbol{n}}_k\big|_{T_1} = k_1 f_k^{T_1} \left[\mu_k \boldsymbol{p}^{\mathrm{B}}(P_{\mathrm{eq}}) - \boldsymbol{n}_k\right], \tag{77}$$

with an optional per-packet profile $f_k^{T_1}$, which allows an orientation-dependent $T_1(\theta)$ [24]. The term is off in the shipped materials ($k_1 = 0$), so no-DNP simulations isolate RF depolarization and internal redistribution.

9 Extended engine: site-resolved populations and the dipolar reservoir

`ExtendedCore` is used when `site_resolved` or `dipolar_enabled` is set. It requires spectral RF, no orientation correlation and no kernel cutoff. With one site and $\beta = 0$ it reproduces the basic equations and their adjoint exactly, which is tested.

9.1 Site resolution

The state becomes $n_{s,k,m}$, with $s \in \{\mathrm{CD}, \mathrm{OD}\}$, and capacities

$$\mu_{s,k} = \mu_k \frac{a_s(x_k)}{\sum_{s'} a_{s'}(x_k)}, \tag{78}$$

where a_s is the site absorption in equation (44). Each site has its own orientation map, so it has its own profiles $f^X_{s,k}$ (equation (57) without the site sum) and its own RF operator. Exchange between sites is scaled by a 2×2 mixing matrix M with off-diagonal elements `exchange_intersite_factor`:

$$(\mathcal{H}V)_s = \sqrt{f_s} \sum_t M_{st}\, S * (\sqrt{f_t}\, V_t). \tag{79}$$

The public state is the site sum $n_k = \sum_s n_{s,k}$. If n is edited externally, it is redistributed over sites with equal level fractions. AFP acts on each site exactly.

9.2 Dipolar reservoir in the high-temperature approximation

The secular dipolar Hamiltonian $\hat{H}_D$ is treated as a separate thermodynamic reservoir with inverse temperature β, in $1/R$ units [8, 9, 11]. To first order in β,

$$E_d = \langle \hat{H}_D \rangle = -C_d\, \beta, \qquad C_d = c_d \sum_{s,k} \mu_{s,k} x_k^2, \tag{80}$$

where c_d is `dipolar_capacity_ratio`; it relates the dipolar heat capacity to the second moment of the offset distribution. Consider a process that takes energy ϵ_{in} from the reservoir. Detailed balance requires $W_{\mathrm{f}}/W_{\mathrm{b}} = e^{-\beta\epsilon_{\mathrm{in}}}$. Expanding to first order, with $W_{\mathrm{f,b}} = W(1 \mp \frac{1}{2}\beta\epsilon_{\mathrm{in}})$, gives the linearized current

$$J = W\Big[(n_a - n_b) - \beta\, \epsilon_{\mathrm{in}}\, \tfrac{1}{2}(n_a + n_b)\Big]. \tag{81}$$

Only the reservoir factor is linearized; the Zeeman populations are not.

RF. A carrier at x_j drives the I^+ transition of packet k. The step $+ \to 0$ raises E_{spin} by x_k and the photon supplies x_j, so the reservoir supplies $x_k - x_j$. Summing over carriers turns the expansion into moments of the kernel:

$$J_k^+ = G_k^+ \Delta_k^+ - \beta H_k^+\, \tfrac{1}{2}(n_{k+} + n_{k0}), \tag{82}$$

$$G_k^+ = c_k \textstyle\sum_j K_{kj} U_j, \qquad H_k^+ = c_k \sum_j K_{kj}(x_k - x_j) U_j, \qquad \mathcal{Q}_k^+ = c_k \sum_j K_{kj}(x_k - x_j)^2 U_j. \tag{83}$$

The energy given to the reservoir is $-\sum_k (H_k^+ \Delta_k^+ - \beta \mathcal{Q}_k^+ \frac{1}{2}(n_{k+} + n_{k0}))$. The I^- terms are analogous, with offset $-x_k - x_j$. The moments are evaluated as $x \circ (KU) - K(xU)$ and $x^2 \circ (KU) - 2x \circ K(xU) + K(x^2 U)$: three convolutions. Off-resonant irradiation cools or heats the reservoir with the Provotorov sign, which is tested.

Flip-flops. Same-branch exchange carries the mismatch $x_i - x_j$:

$$J_{ij}^{(+)} = K_0 S_{ij}\Big[(n_{i0}n_{j+} - n_{i+}n_{j0}) + \tfrac{1}{2}\beta\xi(x_i - x_j)(n_{i0}n_{j+} + n_{i+}n_{j0})\Big], \tag{84}$$

which is evaluated with $x \circ (Sn) - S(xn)$ products. Cross-branch exchange carries $x_i + x_j$. DQ exchange is energy-neutral and gets no β term. ξ (`dipolar_exchange_coupling`) scales the flip-flop part; $\xi = 0$ leaves only the RF–Provotorov coupling.

Reservoir equation.

$$\dot{\beta} = -\frac{\dot{E}_{\text{released}}}{C_d} - r_{1d}\,(\beta - \beta_L), \tag{85}$$

where $\dot{E}_{\text{released}} = -\dot{E}_{\text{spin}}$ from exchange, plus the RF mismatch terms. With exchange alone, $E_{\text{spin}} + E_d$ is conserved *exactly* by the right-hand side, not just to first order; the tests check this to rounding. In the `d_butanol_full_physics` material, $c_d = 0.01$, and $|\beta\delta|$ reaches 0.2–0.7 in slow, power-limited runs. Quantitative use in that regime needs the exponential form of equation (81). For uncapped single-burn optima, site resolution and the reservoir changed Q by ≤ 0.0003.

10 Adiabatic fast passage

An adiabatic frequency sweep across a window inverts every transition resonant in it [19]. The window at x_k contains the I^+ transition of packet k and the I^- transition of packet $\bar{k}$. The sweep is modelled as an instantaneous swap with efficiency ε:

$$(n_{k+}, n_{k0}) \to (\varepsilon n_{k0} + (1-\varepsilon)n_{k+},\ \varepsilon n_{k+} + (1-\varepsilon)n_{k0}), \tag{86}$$

$$(n_{\bar{k}0}, n_{\bar{k}-}) \to (\varepsilon n_{\bar{k}-} + (1-\varepsilon)n_{\bar{k}0},\ \varepsilon n_{\bar{k}0} + (1-\varepsilon)n_{\bar{k}-}). \tag{87}$$

`mirror_safe_indices` keeps at most one index from each mirror pair, so no packet is swapped twice. `qneg_sweep_indices` selects the bins where the signed tensor signal $I^+ - I^-$ is negative (the GUI's "optimal profile" region). The swap conserves μ_k. Its finite duration, and any relaxation during it, are neglected. In the single-burn studies, a two-window AFP reached $Q \approx 0.298$ from $P_0 = 0.50$ at full efficiency.

11 Physical regimes represented by the model

11.1 DNP off, RF on

With $k_{\text{D}} = 0$ and $k_1 = 0$, RF is the only term that changes vector polarization. Exchange feeds population difference into the addressed packets from the rest of the line, while RF equalizes them. The depolarization rate is therefore controlled by four things:

- the RF power and linewidth;
- the two per-spin couplings;
- the two current transition differences;
- the rate at which exchange replenishes those differences.

The replenishment rate depends on the number of partners within the ZQ width ($\propto (S\mu)_k$) and on the DQ partner sum (section 18.7).

11.2 DNP off, RF switched off after a burn

Once RF is off, $\dot{P} = 0$. The exchange channels remove the hole and mirror structure while conserving the reduced P. The line relaxes to the Boltzmann line at $P_{\text{after burn}}$, not the original P_0 (section 7.4). Spectral recovery is not repolarization. The tensor polarization relaxes toward $Q_{\text{B}}(P_{\text{after burn}})$ only as fast as cross-branch exchange allows.

11.3 DNP on, RF off

The line builds toward P_{sat}. Packets have different transient time constants because of w_k^{D}, but the common target fractions give the same asymptotic Boltzmann shape. The transient can be non-Boltzmann across R, because the horns reach the target faster than weak regions. Exchange smooths that difference.

11.4 DNP on, RF on

The system approaches a driven steady state, $\dot{\boldsymbol{n}} = 0$ in equation (50). DNP rebuilds population difference, exchange transports it through packet space, and RF removes it. Hole depth and Q are set by that balance, not by RF power alone. This regime depends most on the model:

- the best steady Q under continuous DNP is 0.20–0.21 with the fitted mixed transport;
- it is 0.24 with same-branch transport only, and 0.30 at hundredfold same-branch exchange;
- the best steady burn sits where only the I^- line exists ($R \approx +1.5$ to $+2.0$), about 15–30 mW;
- burning the horn at the fitted DNP rate is harmful;
- pulsing or bunching the RF in time never beats a steady burn, because incoherent saturation is concave in dose.

11.5 Pulsed operation: optimize, evolve, re-optimize

In practice, ssRF is not run continuously. A pulse is applied, the NMR and control loop take a few seconds to tens of seconds, and the state is measured and re-optimized. After an optimized pulse the optimum changes:

- power moves from the I^--only region to the little peak;
- the pulse becomes shorter and stronger;
- re-applying a fixed program loses up to half of the available gain.

This regime motivates the closed-loop data generator (section 20).

11.6 Negative polarization

All equations are signed. Reversing the spin-temperature polarization reverses the branch signs and the RF currents. For monotone orderings RF still reduces $|P|$. DNP can build toward a negative P_{sat}, and T_1 can relax toward either sign. General non-Boltzmann states require evaluating the currents explicitly, and the optimizer and candidate rule use the signed populations directly.

12 Numerical implementation

12.1 Grid and state initialization

Material	Grid	Step δt
d-butanol materials	$N = 500$, $R \in [-6.186, 6.186]$, $\Delta R = 0.0248$	0.03 s
`demo_single_pake`	$N = 500$, $R \in [-3, 3]$, $\Delta R = 0.0120$	0.03 (arbitrary units)

The initial state is equation (46). A copy is stored as the reference used for plotting direct and mirror changes. It does not enter the dynamics.

12.2 Time integration

The live model integrates equation (50) with positivity-preserving forward Euler and adaptive substeps:

$$\boldsymbol{n} \leftarrow \boldsymbol{n} + h\,\dot{\boldsymbol{n}}(\boldsymbol{n}), \qquad h = \min\Big(t_{\text{event}} - t,\ \frac{0.2}{\lambda_{\max}},\ h_{\text{prog}}\Big). \tag{88}$$

- t_{event} is the next pulse boundary, so every pulse starts and stops exactly.
- h_{prog} is an optional per-program ceiling, applied only while RF is on. The designer sets it to the validated playback step.
- $\lambda_{\max}$ is a conservative bound on the outgoing rate of any population: $W^+ + W^-$, the DNP and T_1 rates, and the exchange bound $2K_0(S\mu)_k + 2r_{\text{x}}K_0|X\mu|_k + r_{\text{DQ}}K_0(C\mu)_k$, evaluated at full occupancy.

For the linear part, $h\lambda \le 1$ keeps every population nonnegative; the factor 0.2 leaves margin for the bilinear terms. Within each material step δt, the step is split as needed. After each substep the code:

1. raises an error if any population is below $-10^{-13}\mu_k$, instead of silently clipping a real error;
2. floors at 10^{-30};
3. rescales each packet to its exact mass μ_k, which removes rounding drift;
4. accumulates the delivered exposure $\sum h\,U_j$ per bin.

Why explicit Euler. The right-hand side costs $O(N \log N)$: three to six FFTs per evaluation. The positivity bound sets the step anyway. And the optimizer needs a forward map whose discrete adjoint is simple and exact (section 14.3). The first-order error is controlled *a posteriori* by step halving (section 14.7). For the fitted d-butanol material the rates are slow: with RF off, the bound allows steps of about 3 s, so the material step $\delta t = 0.03$ s sets the resolution. For the 420 optimized programs of the chain dataset (section 20), at the 0.3 W cap, the validated playback step was 0.015–0.03 s. The rate bound limits the step only for uncapped, high-power commands.

12.3 Pulse programs and event-aware stepping

A program is a list of `BinPulse`(j, U, t_0, τ), each acting on the half-open interval $[t_0, t_0 + \tau)$; profiles add where they overlap. `CompiledProgram` sorts all edges into one event list and rebuilds the command vector $U(t)$ from the pulses active in each interval, instead of adding and subtracting increments. This prevents residual RF after the last pulse. `start_program` anchors program time to the current simulation time, and the integrator splits at every absolute event time. If two boundaries fall below floating-point resolution at a late simulation time, the start is refused, so a pulse is never silently merged.

12.4 Spectral projection and display

The I^+ array is plotted at $R = x_k$. The I^- array is plotted at $R = -x_k$, which is the reversed array on the symmetric grid. The total is their binwise sum. Display noise, when enabled, is added to the plotted arrays at redraw time only; it never feeds back into the populations.

12.5 Conservation summary

- Every packet preserves $\sum_m n_{km} = \mu_k$ after each step.
- Same-branch and DQ exchange conserve N^+, N^0, N^-, and hence P and Q, analytically.
- Cross-branch exchange conserves P and changes Q toward $Q_{\text{B}}(P)$.

- RF conserves packet spin counts and changes P and Q according to equation (58).
- DNP and T_1 conserve packet spin counts and drive P and Q toward their targets.
- With DNP, T_1 and RF off, P is conserved and the state relaxes to the unique Boltzmann line at that P.
- In the extended engine, exchange conserves $E_{\text{spin}} + E_d$.

12.6 Cost

One evaluation of the right-hand side costs:

- one convolution for the RF field;
- one batched three-column FFT convolution for same-branch exchange;
- one more for cross-branch exchange;
- $O(N)$ for DQ, DNP and T_1.

At $N = 500$ this takes tens of microseconds in NumPy. The extended engine multiplies the cost by the number of sites, plus three convolutions per site for the reservoir moments.

13 Snapshots, materials, and reproducibility

A *material* file (`ssrf.material.v1`) stores every kinetic parameter, the optimizer defaults and GUI settings. Older files are upgraded by inserting parameter groups at their neutral defaults, and loading is transactional.

A `PopulationSnapshot` stores:

- the full parameter dictionary;
- n, the reference state, μ and the grid;
- the simulation time;
- for the extended engine, $n_{s,k,m}$ and β.

Its fingerprint is a SHA-256 hash of the kinetic parameters, n, μ and the extended state; display-only settings are excluded. `make_model` rebuilds an `IdealBinModel` and refuses if any of three checks fails:

- the capacities recomputed from the parameters match the stored ones;
- the populations are physical;
- the fingerprint matches.

A snapshot is thus a self-validating initial condition, and every optimizer report records the fingerprint of the state it was computed for.

14 Tensor-polarization optimizer

14.1 Problem

From a frozen state $\boldsymbol{n}(0)$, choose carrier powers $U \in \mathbb{R}^N$, held constant on $[0, T)$ (a *synchronized* program), and a duration T, to solve

$$\max_{U,T} \; Q(\boldsymbol{n}(T)) \quad \text{s.t.} \quad 0 \le U_j \le \bar{U}_j, \quad \sum_j U_j \le P_{\text{cap}}, \quad T \sum_j U_j \le E_{\text{cap}}, \quad T \in [T_{\min}, T_{\max}], \tag{89}$$

with the natural dynamics as captured (DNP, exchange and T_1 on or off). Here P_{cap} is the total-power limit ($\sum U = 1$ corresponds to 0.1 W) and E_{cap} the energy limit ($T \sum U = 34$ corresponds to 3.4 J). The

per-bin ceilings $\bar{U}_j$ are zero outside the candidate set and outside any user regions. At fixed T the two caps merge into one linear constraint,

$$\sum_j U_j \le B(T)/T, \qquad B(T) = \min(E_{\mathrm{cap}},\, P_{\mathrm{cap}} T) \tag{90}$$

(`effective_exposure_limit`). The objective is $Q(T)$ itself. The report always includes the no-RF endpoint, so that natural build-up is never credited to the RF.

14.2 Discretize, then optimize

For a given T, the code fixes a step count K and step $h = T/K$ from a bound that does *not* depend on the trial powers:

$$h = \min\Big(h_{\mathrm{search}},\, \frac{0.18}{\lambda_{\max}\big(\min(\bar{U}, B/T)\big)}\Big) \tag{91}$$

(`prediction_step_plan`). The forward map $\Phi_T : U \mapsto \boldsymbol{n}_K$, with $\boldsymbol{n}_{k+1} = \boldsymbol{n}_k + h\mathcal{F}(\boldsymbol{n}_k; U)$, is then one fixed, smooth function of U. If K depended on U, the objective would be piecewise defined, and its gradient would not exist at the switching points. `FrozenDynamics` holds an algebraically regrouped copy of the right-hand side, which the tests check against `Spin1Model.derivative`.

14.3 Discrete adjoint

Let $\lambda_k = \partial Q/\partial \boldsymbol{n}_k$. Then $\lambda_K = (1, -2, 1)$ in every packet, and

$$\lambda_k = \lambda_{k+1} + h\Big(\frac{\partial \mathcal{F}}{\partial \boldsymbol{n}}\Big|_{\boldsymbol{n}_k}\Big)^{\top} \lambda_{k+1}, \qquad \nabla_U Q = h \sum_{k=0}^{K-1} \Big(\frac{\partial \mathcal{F}}{\partial U}\Big|_{\boldsymbol{n}_k}\Big)^{\top} \lambda_{k+1}. \tag{92}$$

This is the exact gradient of the discrete map [25, 26]. Each term supplies a vector–Jacobian product (VJP).

RF. With $v^+ = \lambda^+ - \lambda^0$ and $v^- = \lambda^0 - \lambda^-$, the state part is $g^+ \mathrel{-}= W^+ v^+$, $g^0 \mathrel{+}= W^+ v^+ - W^- v^-$, $g^- \mathrel{+}= W^- v^-$, and the command part is

$$\nabla_U \mathrel{+}= K^{\top}(c \circ (-\Delta^+) \circ v^+) + \big[K^{\top}(c \circ (-\Delta^-) \circ v^-)\big]^{\mathrm{rev}}. \tag{93}$$

Since K is symmetric, this is one more convolution (`RFRateOperator.transpose`).

Bilinear exchange. Every exchange current has the form $a_i(\mathsf{S}b)_i$ with a symmetric kernel S. Its VJP contributes $\lambda_i(\mathsf{S}b)_i$ to ∂a, and $\mathsf{S}(\lambda \circ a)$ to ∂b. The adjoint therefore uses the same FFTs as the forward pass. The extended engine has the same structure, with the moment kernels $x \circ \mathsf{S} - \mathsf{S} \circ x$ and the scalar adjoint of β carried along.

Cost. The backward pass costs about twice the forward pass. It stores the trajectory ($8 \cdot 3NK$ bytes, capped at 512 MiB). One objective-plus-gradient evaluation therefore costs about three forward integrations, independent of N.

14.4 Candidate bins and seeds

At $t = 0$, equation (58) gives the initial slope

$$\frac{\partial \dot{Q}}{\partial U_j}\Big|_{t=0} = \left[K^\top(-3c \circ \Delta^+) + \left(K^\top(3c \circ \Delta^-)\right)^{\mathrm{rev}}\right]_j. \tag{94}$$

The default candidate set (`union`) is the bins where this slope is positive, together with the bins where the signed tensor signal $I^+ - I^-$ is negative. Other bins get $\bar{U}_j = 0$. The seeds are:

- $\bar{U}$ on the positive-slope bins, for the first duration;
- the previous duration's solution, rescaled to the budget (continuation), for later durations;
- random fractions of $\bar{U}$, for any additional starts.

14.5 Inner solver

- **Budget not binding** ($B/T \geq \sum \bar{U}$): L-BFGS-B [27] on box constraints, in coordinates scaled to $[0, 1]$.
- **Budget binding**: SLSQP [28] with the linear inequality of equation (90). The coordinate scales are capped at ten times the mean available power per bin; otherwise per-bin ceilings far above the budget cause oversized steps. The final point is projected onto the budget and re-evaluated, so an infeasible Q is never reported.

The objective is scaled by 10^3, so that small per-bin gradients are not mistaken for convergence. The code keeps the best *feasible* iterate, not the solver's last point.

14.6 Duration scan and selection

1. Solve on a geometric grid of `duration_samples` values in $[T_{\min}, T_{\max}]$.
2. Refine `duration_refinements` times. Each refinement places $\sqrt{T_a T_b}$ in the wider, in $\log T$, of the two intervals next to the best horizon. Refining both sides avoids missing an interior optimum.
3. Re-evaluate every completed horizon at the finer endpoint step $\min(\delta t, h)$.
4. Select the *shortest* T whose Q is within $\tau = \max(\tau_{\mathrm{abs}}, \tau_{\mathrm{rel}}(Q_{\mathrm{best}} - Q_0))$ of the best; the defaults are 10^{-6} and 0.2% of the gain. Shorter pulses deposit less energy and are less sensitive to model error.
5. If no horizon beats Q_0, return "no RF". The designer never proposes irradiation that lowers Q.

14.7 Validation, controls, and diagnostics

- **Replay.** The selected program is turned into a real `PulseProgram` and replayed through the unchanged scheduler and model.
- **Half-step check.** The replay endpoint is compared with an evaluation at half the step. For first-order Euler the full-step error is about twice that difference. The playback step is halved, up to `max_validation_halvings` times, until $2|\Delta Q| \leq \epsilon$ (2×10^{-5} by default) and the state difference meets 10ϵ. Otherwise the status is `needs_finer_time_step`. The validated step is exported as `recommended_playback_dt`.
- **Control.** The no-RF endpoint at the same T is reported, so the RF advantage $Q - Q_{\mathrm{noRF}}$ is explicit.
- **First-order gap.** At the solution, the linearized problem $\max_{u \in \mathcal{U}} \nabla Q \cdot (u - U)$ over the box plus budget is a continuous knapsack. It is solved exactly by sorting the gradient. Its value bounds the gain still available to first order. It is a local KKT diagnostic, not a global certificate.

- **Optional smoothing.** Gaussian smoothing of U (preserving total power and the mask) is accepted only if, after the same replay and time-step validation, it loses no more Q than the selection tolerance that remains unspent.

Scope. This is local search over *synchronized* programs from a few seeds, with no global certificate:

- time-shaped programs (4–8 segments) gained ≤ 0.0025 in Q;
- periodic schedules under continuous DNP gained ≤ 0.0016;
- the search has a wall-time budget; when it runs out, the best validated horizon is returned with a warning.

15 Software architecture and code-to-physics mapping

Paths are under unless they start at the repository root (marked /).

File	Role
`/run_app.py`	GUI entry point; takes a material name or file, `-program`, `-state`, `-list`.
`/design_tensor.py`	Headless tensor design from a material or snapshot.
`/generate_training_data.py`	Command-line dataset generator (`generate`, `info`, `export`).
`lineshape.py`	Boltzmann relations, $P, Q \leftrightarrow$ level fractions, analytic single-site Pake branch.
`multisite_lineshape.py`	Azimuth-averaged complex powder branch; C–D/O–D mixture.
`model.py`	`Spin1Params`, `Spin1Model`: grid, capacities, observables, mechanism profiles, pair kernels, full right-hand side.
`spectral_rf.py`	`RFRateOperator`: $U \mapsto W^{\pm}$ and its transpose.
`ideal_model.py`	`IdealBinModel`: per-bin programs, event-aware adaptive stepping, extended-engine switch.
`extended_core.py`	Site-resolved and dipolar-reservoir dynamics with exact VJP.
`afp.py`	AFP swap and region selection.
`pulse_program.py`	Bin-addressed schedules, JSON/CSV I/O, compilation to an event list.
`material_config.py`	Versioned material files and fingerprinted population snapshots.
`presets.py`	Shipped material names and paths.
`tensor_optimizer.py`	`FrozenDynamics`, adjoint, `optimize_tensor`, replay and validation.
`datagen.py`	Random, sweep and chain dataset generation; shard I/O.
`playback.py`	RF gate and program-start orchestration, independent of Qt.
`baseline_gui.py`, `gui.py`, `designer_gui.py`	Qt windows (layered by subclassing).
`rf_response_gui.py`	RF saturation / spin-diffusion parameter dialog.
`profile_editor.py`, `profile_editing.py`	Per-bin program table editor.
`/tools/steady_state_rf_dnp.py`	Steady-state RF + DNP optimizer.
`/tools/plot_tensor_design.py`	Plots of a design result.
`/ml/`	PyTorch pulse-design and polarization models (section 21).
`/materials/, programs/`	Shipped materials and reference programs.
`/tests/`	460 regression tests (section 17).

15.1 Core function map

Table 1: Key functions and their mathematical role.

Function	Meaning
`normalized_component`	Normalized analytic single-site branch, equation (42).
`powder_branch, site_components`	Resolvent equation (43) and the two-site density equation (44).
`level_populations_from_PQ`	equations (17) to (19), with a check of the physical triangle.
`Spin1Model.reset`	Grid, μ_k, initial state, reference copy, display calibration.
`polarizations, branch_areas`	P, Q, $Q_{\mathrm{B}}(P)$; branch areas.
`spectrum_from_state`	Projection to $I^{+}(R)$, $I^{-}(R)$ and total, equations (47) and (48).
`branch_indices`	Physical R to packets k_{+} and k_{-}, equations (33) and (34).
`packet_site_geometry`	O–D share and per-site $\cos^2\theta$ of each packet.
`mechanism_profile`	f_k^X, equation (57).
`rf_packet_response`	c_k, equation (56).
`rf_rate_operator`	Builds `RFRateOperator` with κ, Γ_k, c_k.
`capacity_rate_weights, dnp_rate_weights`	DNP density weight w_k^{D}, equation (75).
`_spectral_overlap`	$S(\delta)$, equation (66).
`_ensure_diffusion_kernel`	Builds and caches the Toeplitz/Hankel FFT kernels or sparse pair lists.
`_spin_diffusion_terms`	Same-branch, cross-branch and DQ currents, equations (67) to (69) and (71).
`derivative`	Full right-hand side; optional breakdown by mechanism (section A.11).
`IdealBinModel.step`	Event-aware adaptive Euler, equation (88); positivity check; exposure accounting.
`_max_outgoing_bound`	$\lambda_{\max}$ for step selection.
`set_program, start_program, stop_program`	Install, anchor and stop a pulse program.
`afp_sweep, afp_target_state`	AFP swap (section 10), per site in the extended engine.
`ExtendedCore.rhs / vjp`	Extended dynamics and its exact adjoint.
`PopulationSnapshot.capture / make_model`	Fingerprinted state freeze and validated restore.
`FrozenDynamics.rhs / vjp / evaluate`	Regrouped forward map, VJP, and fixed-step forward and adjoint integration.
`optimize_tensor`	Duration scan, inner solves, selection, validation, report.
`replay`	Independent prediction through the unchanged scheduler.
`synchronized_program`	(U, T) to a `PulseProgram`.
`datagen.make_chain, undershoot`	Closed-loop data generation (section 20).

15.2 GUI update loop

The real-time loop is

Listing 1: Simplified GUI update logic.

```
every 35 ms (QTimer), if not paused:
    model.step(n_steps=steps_per_tick) # adaptive, event-split, exact pulse edges
    record I+(M,t), I-(M,t), I+(-M,t), I-(-M,t)
    if program finished and "pause at end": pause
update spectrum (+ RF overlay), tensor spectrum, monitor traces, readouts
```

The timer decides only when to draw. Pulse durations are simulation time, independent of the redraw rate. `playback.py` keeps two RF conditions separate: an enabled RF gate and a started program. Loading a file arms neither.

16 Graphical interface

`run_app.py` opens the main window, *ssRF — spin-1 tensor-polarization simulator*, paused with RF off, at the chosen material (default `d_butanol_matched`).

16.1 Three dynamic plots

1. **Spectrum.** $I^+(R)$, $I^-(R)$ and their sum, with the monitor marker M and its mirror $-M$. An optional overlay shows the RF commands (`Show RF commands on spectrum`).
2. **Tensor spectrum.** $I^+(R) - I^-(R)$. Its integral is Q, and its negative regions are where RF raises Q (section 6.5).
3. **Monitor traces.** I^+ and I^- at M (solid) and at $-M$ (dotted), against time since the trace was started.

16.2 Live readouts

$P(t)$ and $Q(t)$, $Q_{\mathrm{B}}[P(t)]$, simulation time, RF/DNP/diffusion states, program state and elapsed time, delivered exposure, and the material name and RF-response summary.

16.3 Controls, defaults, and ranges

Defaults are those of `d_butanol_matched`, the material the GUI opens with. The designer window widens any range to include a value loaded from a material file. For example, the zero-quantum width 0.717 exceeds the nominal maximum of 0.5.

Table 2: GUI controls, defaults (`d_butanol_matched`), ranges, and meaning.

Group	Control	Default	GUI range	Meaning
Material	load / save / save as	matched	files	Whole material (parameters, optimizer, GUI settings).
Material	edit ALL settings	—	dialog	Every `Spin1Params` and optimizer field, validated.

Group	Control	Default	GUI range	Meaning
Material	RF saturation / spin diffusion	—	dialog	κ, Γ, selectivity band, exchange parameters (sections 6.4 and 7.2).
Material	save / load state	—	files	Fingerprinted population snapshot (section 13).
Material	generate tensor program	—	dialog	Captures the current state and opens the designer.
Run	RF ON/OFF	off	toggle	Gate for the RF term; simulation time continues.
Run	pause / reset / restart traces	running, paused at start	buttons	Stop stepping; rebuild the initial state; clear the monitor traces.
Markers	monitor bin M	bin nearest $R = -0.935$	$[0, 499]$	Packet monitored by the solid traces; markers never move RF.
Markers	monitor R	-0.935	$[R_{\min}, R_{\max}]$, step ΔR	Snaps to the nearest bin; mirror $-M$ is read-only.
Mode	manipulation mode	ss-RF only	ss-RF / AFP / both	Which panels are active.
RF program	global RF gain	1.0	$[0, 1000]$	Multiplies every scheduled command.
RF program	bin table	empty	per bin	Power U, start, duration of every bin.
RF program	start / stop / pause at end	—	buttons	Anchor the program at the current time; stop while evolving.
RF program	show RF commands	off	toggle	Overlay of U on the spectrum.
AFP	region	optimal ($Q < 0$)	optimal / selective	Bins with $I^+ - I^- < 0$, or an $[L, R]$ window.
AFP	L, R	-0.98, -0.72	$[R_{\min}, R_{\max}]$	Selective window bounds.
AFP	efficiency ε	1.0	$[0, 1]$	Swap efficiency.
DNP	DNP ON/OFF	off	toggle	DNP source term.
DNP	P saturation	0.476	$[-0.99, 0.99]$	P_{sat}.
DNP	DNP build rate	0.00230	$[0, 20]$	k_{D} [s^{-1}].
Lineshape	Γ broadening	0.0236	$[0.001, 0.5]$	Line width; resets the model.
Lineshape	$\eta \cos 2\phi$	0	$[-0.5, 0.5]$	Single-site asymmetry (unused for the two-site line).
Lineshape	plot scale	0.102	$[0.01, 20]$	Display only.
DNP weighting	rate-density power α_{D}	1.0	$[0, 3]$	DNP density weighting.

Group	Control	Default	GUI range	Meaning
DNP weighting	max weight clip c	12	$[1, 100]$	Clip of the raw weights.
Spin diffusion	diffusion ON/OFF	on	toggle	All exchange channels.
Spin diffusion	exchange scale K_0	0.01283	$[0, 1000]$	Flip-flop rate scale.
Spin diffusion	ZQ overlap shape	Lorentzian	Lor. / Gauss.	$S(\delta)$.
Spin diffusion	zero-Q width w	0.717	$[0.001, 0.5]^*$	ZQ overlap HWHM.
Spin diffusion	tensor exchange ratio	1.0	$[0, 10]$	r_{x}.
Spin diffusion	DQ transport ratio	3.03	$[0, 10]$	r_{DQ}.
Spin diffusion	orientation corr.	0	$[0, 1]$	f_θ.
Spin diffusion	corr. width	20°	$[0.1, 90]$	σ_θ.
Spin diffusion	cutoff	0 (all tails)	$[0, 100]$	In units of w.
Spin diffusion	MW diffusion factor	1.0	$[0, 20]$	Multiplies K_0 while DNP is on.
Initial / numeric	initial P	0.50	$[-0.99, 0.99]$	Boltzmann start; resets the model.
Initial / numeric	T_1 rate, P_{eq}	0, 0	$[0, 10]$, $[-0.99, 0.99]$	Spin-lattice relaxation.
Initial / numeric	dt	0.03 s	$[10^{-5}, 0.05]$	Material step; substeps are automatic.
Initial / numeric	steps/tick	12	$[1, 1000]$	Material steps per 35 ms timer tick (0.36 s of simulation per tick).
Initial / numeric	plot noise	0	$[0, 0.02]$	Display-only noise.

*Widened automatically to include the loaded value.

16.4 Backend-only parameters

The following are stored in the material file but have no main-window control:

- the grid (`n_bins`=500, `r_min`/`r_max`=$\mp$6.186) and the optional initial `q0`;
- the two-site lineshape parameters and `line_phi_points`=128;
- all mechanism-profile parameters (equation (57)) and the extended-engine switches (site resolution, dipolar reservoir);
- the legacy fields (`gamma_rf`=2.0, Voigt widths, `rf_density_power`, `rf_pedestal_gain`), which are unused in spectral mode;
- the display settings (`display_scale`=0.102, `plot_divisor`=10, `calibration_p`=0.5);
- the GUI settings: timer 35 ms, 12 steps per tick, 4500 trace points, window 1200 × 555.

16.5 Tensor designer window

Generate tensor program from CURRENT state pauses the simulation and captures a snapshot. It opens the designer, which has the following settings:

- maximum command per bin;
- duration range and number of samples;
- iterations per start;
- wall-time limit;
- maximum prediction steps;
- total power limit and total exposure limit (each optional);
- profile smoothing;
- candidate rule: union, tensor deficit, positive slope, or all bins;
- exact/advanced JSON: regions, per-bin ceilings, starts, tolerances.

The calculation runs in a worker thread; cancelling leaves the live model unchanged. Three tabs show the progress and validation log, the predicted endpoint with the RF profile, and the bin command table. *Load generated program (do not run)* installs the program with its validated playback step as the RF integration ceiling. Export writes JSON, CSV and the snapshot.

17 Validation and regression tests

The suite contains 460 tests (`python3 -m pytest`). The GUI tests skip without a display. They establish implementation consistency and the physics invariants, not experimental validity.

Table 3: Regression-test coverage (number of tests in parentheses).

File	Assertions
test_tensor_designer (48)	Regrouped RHS equals the model; analytic VJP and discrete adjoint match finite differences; designs bounded, improving and replayable; refuses damaging RF; negative P; cancellation leaves the live state alone; single-bin design matches the analytic optimum exposure.
test_live_diffusion_fft (38)	FFT, sparse and pair-sum evaluators agree; every channel conserves μ_k and P over random grids, overlaps, cutoffs and cross ratios.
test_boltzmann_recovery (30)	Each channel stationary at Boltzmann; positive entropy production; non-Boltzmann uniform states not stationary; relaxation to the unique Boltzmann line at the remaining P; DQ leaves Q unchanged; RF stays an uncancelled sink during recovery; dt convergence.
test_model_diffusion_kernel (21)	Ideal-bin mirror relation emerges from the populations; packet mass and P conservation; cross channel conserves P and changes Q; post-burn currents fill holes and reduce mirror peaks; finite RF width imposes no fixed mirror ratio.
test_ideal_profiles (31), test_custom_editing (32), test_playback (17), test_rf_integration_clock (11)	Program editing and compilation; exact pulse edges; RF gate and program state; integration ceiling only during RF; negative P with simultaneous mirror drive.

File	Assertions
test_spectral_saturation (18), test_peak_selectivity (11), test_material_response (17)	Spectral kernel and its transpose; selectivity and pedestal profiles; legacy and independent modes reproduce earlier laws.
test_v8_mechanisms (10), test_v8_extended (9)	Mechanism profiles; one-site extended engine equals the basic engine; extended adjoint at $\beta \neq 0$; exchange conserves $E_{\text{spin}} + E_d$; Provotorov sign; per-site AFP.
test_multisite_material (17), test_production_materials (12)	Two-site lineshape; the shipped materials load, are fingerprinted and reproduce their reference values.
test_exposure_budget (16), test_total_power (12), test_prediction_step_limits (6), test_profile_refinement (11)	Caps never exceeded; SLSQP path; step plans independent of trial powers; smoothing accepted only within tolerance.
test_afp (9), test_material_afp (30)	Mirror-safe windows; $Q < 0$ region selection; efficiency; per-site action.
test_datagen (7)	Determinism; shard/manifest/resume/export round trip; late-snapshot replay; states-only; chain rounds follow applied programs; undershoot respects caps.
GUI tests (47)	Display projections, callbacks and dialogs (headless where possible).

18 How to calibrate the simulation software against experiment

Quantitative use requires a staged calibration, not simultaneous free fitting of every parameter. The column "d-butanol" states what was done for the shipped materials.

18.1 Step 1: static lineshape and display calibration

With RF and DNP transients absent, fit the unmanipulated spectrum for:

- the frequency-to-R conversion and centre;
- the two-site parameters γ, ρ_{OD}, f_{OD}, η_{CD}, η_{OD} (or Γ_{line} and s for one site);
- baseline, phase, gain and the absolute NMR calibration C.

A real Q-meter fit may also require dispersion, cable response and a complex-susceptibility treatment [23, 24]. *d-butanol:* fitted to 2014 spectra, held-out RMS 0.79%.

18.2 Step 2: initial population state

Use calibrated branch areas for P_0 and, where possible, Q_0. Use $Q_0 = Q_{\text{B}}(P_0)$ only if one spin temperature is verified. Otherwise load a non-Boltzmann state as a snapshot.

18.3 Step 3: DNP-only build-up

With RF off, fit P_{sat} from the asymptote, k_{D} from the time scale, and the DNP density weighting (α_{D} or f^{dnp}) from horn-versus-shoulder build-up rates. These are now DNP-only parameters, so a DNP fit no longer constrains RF or recovery. *d-butanol:* $k_{\text{D}} = 2.30 \times 10^{-3}\,\text{s}^{-1}$, $P_{\text{sat}} = 0.476$, held-out 0.79%.

18.4 Step 4: short, low-power RF calibration

Use burns short enough that exchange is negligible during the pulse:

- κ from the initial fractional depletion rate;
- Γ from the narrow-hole width;
- per-spin selectivity (s, f^{rf}, f^{width}) from the initial fractional rates at horn and pedestal;
- the direct/mirror ratio at small $|R|$ (section 6.6).

d-butanol: κ was obtained jointly with exchange from one 0.1 W burn/refill experiment (held-out RMSE 3.4×10^{-4}). Selectivity was not measured; $s = 0.7$ is a choice.

18.5 Step 5: RF-off recovery

After a calibrated burn, switch RF and microwaves off and measure:

- the initial refill slope at the burn bin and its width dependence;
- mirror-peak decay at $-R$;
- the evolution of neighbouring bins;
- $P(t)$, to confirm redistribution without repolarization;
- $Q(t)$, whose relaxation toward $Q_{\mathrm{B}}(P)$ measures the cross-branch channel.

Fit K_0, w, r_{x} and r_{DQ}. A narrow burn in the I^--only region ($R \approx 1.8$) separates the channels: same-branch transport refills only I^- there, while cross-branch transport also changes I^+ at the mirror frequency. *d-butanol:* $K_0 = 0.0128$, $w = 0.717$, $r_{\mathrm{DQ}} = 3.03$ from the same burn/refill fit; $r_{\mathrm{x}} = 1$ assumed. A narrow physical width ($w = 0.047$) fits 23% worse.

18.6 Step 6: DNP-on competition

Run DNP and RF together. The calibrated terms should predict the steady hole depth, P and Q. A gentle burn at $R \approx +1.9$ with DNP on, watching Q rise above its DNP-only value and approach steady state over $\sim$435 s, is the most direct test of the transport assumption. *d-butanol:* not yet done.

18.7 Saturation balance: peak versus pedestal

The motivating observation is that, in deuterated butanol, a hole can need less RF power in the pedestal region near $\theta \simeq 0$ than at a Pake peak near $\theta \simeq 90°$. This section derives what the model says about that difference and evaluates it with the calibrated materials. Earlier spin-1 studies provide the RF/DNP and recovery context [21–24].

Physical basis. Khutsishvili's magnetization equation contains distinct diffusion, RF-saturation and relaxation terms. Its stationary solution shows the competition between saturation and restoration [14, Eqs. (3) and (12)]. Walder *et al.* measured the refilling of controlled proton-magnetization holes by spin diffusion [18]. Those holes are spatial deficits in hydrated cementitious solids, so the experiment establishes a transport mechanism and a recovery-based measurement strategy, not a deuteron peak–pedestal dependence. For deuterium, Eckman explains why matching transitions within a local-field width permits first-order dipolar exchange, and why quadrupolar mismatch suppresses it [17, Sec. III.G]. Suter and Ernst distinguish the transport of Zeeman and quadrupolar order in resolved spectra [16].

A local balance. Use the population difference per spin in packet k,

$$d_{k,+} = \frac{n_{k+} - n_{k0}}{\mu_k}, \qquad d_{k,-} = \frac{n_{k0} - n_{k-}}{\mu_k}. \tag{95}$$

For an isolated addressed transition, write either difference as d_k. A locally quasi-steady approximation is

$$\dot{d}_k = -\lambda_{\mathrm{RF},k} d_k + \kappa_k (d_{\mathrm{res},k} - d_k), \qquad \lambda_{\mathrm{RF},k} = 2W_k, \tag{96}$$

where κ_k is the effective restoring rate and $d_{\mathrm{res},k}$ the reservoir difference. For fixed coefficients,

$$\frac{d_{k,\mathrm{ss}}}{d_{\mathrm{res},k}} = \frac{\kappa_k}{\lambda_{\mathrm{RF},k} + \kappa_k}, \tag{97}$$

$$d_k(t) = d_{k,\mathrm{ss}} + \left[d_k(0) - d_{k,\mathrm{ss}}\right] \exp[-(\lambda_{\mathrm{RF},k} + \kappa_k)t]\,. \tag{98}$$

Demanding a residual fraction ϵ gives

$$\lambda_{\mathrm{RF},k}^{\mathrm{required}} = \frac{1-\epsilon}{\epsilon}\,\kappa_k, \qquad U_k^{\mathrm{required}} = \frac{1-\epsilon}{\epsilon}\,\frac{\kappa_k}{\lambda_{\mathrm{RF},k}/U}. \tag{99}$$

Stronger restoration requires more RF for equal fractional saturation. The packet capacity cancels from this criterion; it still sets the absolute signal and the absorbed power. With several reservoirs (exchange with unburned spins, DNP, lattice),

$$d_{k,\mathrm{ss}} = \frac{\sum_a \kappa_{ka} d_{ka}^{\mathrm{res}}}{\lambda_{\mathrm{RF},k} + \sum_a \kappa_{ka}}. \tag{100}$$

With DNP off, T_1 does not restore the hyperpolarized state, and a sustained burn depletes a finite unpumped reservoir. In that case d_{res} is not constant.

Converting a saturation rate to RF amplitude. With equation (51), $\lambda_{\mathrm{RF},k} = 2\omega_1^2\Gamma_k/(\Gamma_k^2 + \Delta_k^2) \equiv c_k' B_1^2$, and

$$B_{1,k,\mathrm{required}}^2 = \frac{1-\epsilon}{\epsilon}\,\frac{\kappa_k}{c_k'}. \tag{101}$$

A larger required field at the peak therefore constrains the ratio κ_k/c_k'. It identifies faster replenishment only once the per-spin RF response c_k' is controlled separately. Here Γ_k is the homogeneous packet width. It is neither the Pake-envelope width nor the ZQ width w.

Replenishment from the pair currents. The production model gives κ_k explicitly. Take a small I^+ hole in packet k, $p_{k+} = \bar{p}_+ - \delta$, $p_{k0} = \bar{p}_0 + \delta$, with all other packets at the Boltzmann fractions $\bar{p}$. Linearizing equations (67) and (68) gives

$$\kappa_k^{\mathrm{SQ}} = K_0\left[(S\mu)_k - \mu_k\right]\left(1 - \frac{\bar{p}_-}{2}\right), \qquad (S\mu)_k = \sum_j S(x_k - x_j)\,\mu_j. \tag{102}$$

The same-branch refill rate is proportional to the *number of partners within the ZQ width*, $(S\mu)_k - \mu_k$. The cross-branch rate is proportional to the partners at the opposite frequency, $(X\mu)_k$. The DQ rate is proportional to the global partner sum, which is the same for every packet. For a narrow ZQ width, $(S\mu)_k$ follows the local Pake density, and a horn packet has many more partners than a pedestal packet. For a broad width it is nearly uniform. The powder density thus enters replenishment through partner counting at a specified width. That is a definite, testable mechanism, not an imposed weight.

Calibrated predictions. Table 4 evaluates κ_k (from the full model, linearized) and the per-spin RF rate for a narrow I^+ hole. The state is Boltzmann at $P = 0.45$ with DNP off. The peak is at $R = 0.93$; the pedestal at $R = -1.60$, on the same branch toward $\theta = 0$.

Three conclusions follow:

Material	κ_k [s^{-1}]		λ_{RF}/U [s^{-1}]		U^{req} ratio	Peak channels
	peak	pedestal	peak	pedestal	peak/ped.	SQ / X / DQ
matched ($w = 0.717$, $r_{DQ} = 3.03$)	0.0114	0.0079	2.61	2.61	1.4	0.0061 / 0.0028 / 0.0026
same-branch ($w = 0.047$, SQ only)	0.0435	0.0101	1.44	1.44	4.3	0.0435 / 0 / 0
extend (horn-favouring RF)	0.0114	0.0079	14.2	0.64	0.065	as matched

Table 4: Peak–pedestal saturation balance from the calibrated materials, equation (99). U^{req} is the command needed for a given fractional saturation.

1. With equal per-spin RF rates (the matched and same-branch materials), the peak needs *more* RF than the pedestal. The ratio is mild (1.4) with the fitted broad, nonlocal transport and large (4.3) with narrow, local same-branch transport. The observed contrast is thus naturally explained by *partner counting at a narrow ZQ width*.
2. Horn-favouring per-spin RF selectivity (`d_butanol_extend`) predicts the *opposite*: the peak needs about 15 times less RF. That material maximizes single-burn Q, but it is disfavoured if the peak–pedestal observation holds.
3. The observation therefore discriminates between two unconstrained parameters that matter most for tensor enhancement: the per-spin selectivity and the locality of transport. A quantitative peak–pedestal measurement (required power at matched fractional depth, plus initial refill rates) should be the next calibration experiment.

These are linearized, single-bin rates with an unburned reservoir. A finite hole width depletes the nearest partners too, which lowers the effective κ. This happens most for narrow w, so the full-model required power should be computed for the actual burn width (`tools/steady_state_rf_dnp.py` and the GUI support this directly).

Controlled simulation comparisons.

1. **Compare the same transition at its peak and pedestal.** Compare I^+ near $R = +1$ with its pedestal toward $R = -2$, or I^- near $R = -1$ with its pedestal toward $R = +2$. Comparing I^+ and I^- at one R tests overlap selectivity instead. Hold duration, bandwidth and calibrated drive fixed, and compare fractional depths. For a required-drive comparison, fix the target depth and vary the drive.
2. **Capacity-only baseline.** With all profiles at 1 and exchange, DNP and T_1 off, $d_k(t)/d_k(0) = \exp(-2W_k t)$ is the same at every location despite unequal absolute signals. This is the null check.
3. **Separate RF response from transport.** Vary f^{rf} or f^{width} with exchange off, then vary K_0, w and the channel ratios with RF fixed. Use the derivative breakdown (section A.11) to record each channel's contribution.
4. **Unpumped versus maintained reservoirs.** Compare burns and dark recovery with DNP off (retaining the reduced P) and with DNP on, after verifying the driven steady state.
5. **Time, bandwidth and acquisition.** Compare early depletion, late depth and recovery for matched burns. Vary the burn width, and include readout delay or sweep averaging when comparing with data. Under grid refinement, preserve the physical bandwidth and the total command.

19 Calibrated materials

The other shipped materials differ only where the data are silent.

Quantity	`d_butanol_matched`	Source
Grid	$N = 500$, $\lvert R\rvert \le 6.186$	covers the O–D line
Line	two-site, $\gamma = 0.0236$, $\rho_{\mathrm{OD}} = 1.355$, $f_{\mathrm{OD}} = 0.021$ $\eta_{\mathrm{CD}} = 0.065$, $\eta_{\mathrm{OD}} = 0.033$	2014 spectra, held-out RMS 0.79%
RF	$\kappa = 0.0196\,\mathrm{s^{-1}U^{-1}}$, $\Gamma = 0.015$	0.1 W burn/refill, RMSE 3.4×10^{-4}
Selectivity	$s = 0.7$ at $R_c = 0.935$ (half-width 0.16, taper 0.25)	chosen, not fitted
Exchange	$K_0 = 0.0128$, Lorentzian $w = 0.717$ $r_{\mathrm{x}} = 1$, $r_{\mathrm{DQ}} = 3.03$	same burn/refill fit r_{x} never fitted
DNP	$k_{\mathrm{D}} = 2.30 \times 10^{-3}\,\mathrm{s^{-1}}$, $P_{\mathrm{sat}} = 0.476$, $\alpha_{\mathrm{D}} = 1$	2014 build-up, held-out 0.79%
Step	$\delta t = 0.03\,\mathrm{s}$	
Units	time in seconds; $\sum U = 1$ is 0.1 W	event IDs read as seconds

Table 5: The experimental fit.

Material	Differences from `matched`	Best single-burn Q from $P_0 = 0.50$
`d_butanol_matched`	—	0.245–0.248
`d_butanol_basic`	$s = 1$; RF coupling density exponent 0.5; RF width $a_2 = 0.6$	0.265
`d_butanol_extend`	$s = 1$; coupling exponent 1; width density exponent -0.3; $a_2 = 0.6$	0.275
`d_butanol_full_physics`	site resolution and dipolar reservoir ($c_d = 0.01$)	as matched ± 0.0003
`d_butanol_same_branch_transport`	hypothesis: narrow width $w = 2\Gamma = 0.047$; refit $K_0 = 0.516$, $\kappa = 0.0108$, $k_{\mathrm{D}} = 2.28 \times 10^{-3}$, $P_{\mathrm{sat}} = 0.451$; $r_{\mathrm{x}} = 0$, $r_{\mathrm{DQ}} = 0$	steady RF+DNP 0.241
`demo_single_pake`	single site on $[-3, 3]$; uncalibrated	—

The largest model levers on tensor enhancement are:

- per-spin RF selectivity, about 0.04 in single-burn Q (unidentified);
- the type and strength of transport under continuous DNP, up to 0.09 (weakly identified);
- the RF law, 0.015–0.02 (the density double counting of earlier versions).

Calibration, the power cap and the lineshape each move Q by less than 0.01. The repository page on key results gives the full list.

20 Command-line training-data generation

`generate_training_data.py` turns the simulator and optimizer into a deterministic dataset factory for machine learning. Each sample pairs an input with a target:

- **Input:** the total spectrum of the input state (section 3), optionally with the branch signals and level populations.
- **Target:** the optimizer's (U, T), with its predicted P and Q, the no-RF control, the validated playback step, and optionally the post-pulse evolution.

20.1 State distributions

- **Random mode** (`-samples`). A Boltzmann start at random P_0, then a random history: log-uniform burn windows, powers and durations; optional AFP; free evolution with DNP on, off or random.
- **Sweep mode** (`-p0-values`). A grid of P_0 × random burns × fixed evolution times. All times of one burn share its history, and the time since the burn is a model input.
- **Chain mode** (`-chains`, closed loop). Optimize from the current state (one sample). Apply an *undershot* version of that program. Evolve. Repeat for `-rounds` rounds.

The undershoot in chain mode works as follows:

- the power is scaled by $a \sim \mathcal{U}[0.3, 1]$ and the duration by $b \sim \mathcal{U}[0.2, 1]$;
- the per-bin profile gets log-normal noise ($\sigma = 0.5$), is renormalized, and is clipped to the program's own maximum;
- with probability 0.15 the exact optimum is applied instead;
- the evolution time is log-uniform in 2–40 s, the NMR and control-loop latency.

Undershooting only lowers power and time, so every applied pulse satisfies the caps. The time since the last pulse is recorded as a model input.

20.2 Why closed-loop data

In a 240-sample sweep pilot (P_0 25–35%, DNP on, 0.3 W cap), the optimal profile barely depended on the state:

- the cosine similarity of every target to the mean was at least 0.982;
- a single fixed program reached 99% of the optimizer's gain.

In chain data the optimum changes from round to round:

Round	Q before pulse	Q with optimum	Mean duration	Power below $R = 0$
0 (fresh)	0.068	0.125	60 s (bound)	29%
1	0.114	0.130	28 s	43%
2	0.119	0.132	20 s	46%
3	0.124	0.133	14 s	46%

Re-applying the pilot's mean program on later-round states gave 75% of the gain on average, and sometimes ended with lower Q than no RF.

20.3 Implementation

- **Determinism.** Every random draw comes from `np.random.default_rng([seed, tag, unit, k])`, so any sample can be regenerated alone.
- **Shards.** Work units (samples, sweep groups or chains) are written as `.npz` shards through a temporary file and an atomic rename, so a shard is either complete or absent.
- **Manifest and resume.** A manifest stores the configuration, the full material files and the grid. A run resumes only if its configuration is identical.
- **Parallelism.** Workers are spawned processes with one BLAS thread each.
- **Optimizer presets** bound the wall time. The `fast` preset uses 5 durations, 1 refinement, 40 iterations and 1 start; at the 0.3 W cap one design takes 1–3 CPU-minutes, about 200 samples per hour on 7 workers.

- **States only** (`-states-only`) skips the optimizer and stores spectra with exact P, Q and branch labels, for polarization models, at about 0.5 CPU-s per sample.

21 Learning models

21.1 Pulse-design model

`SpectrumToPulse` is a 1D U-Net over the frequency axis:

- **Encoder:** three down-sampling stages to $N/8$; convolutions with kernel 5, GroupNorm and GELU.
- **Decoder:** skip connections; a 1×1 convolution gives profile logits ℓ.
- **Scalar head:** an MLP on the mean- and max-pooled bottleneck gives the total power and $\log T$.
- **Inputs:**
  - the spectrum with one *global* scale, since the absolute amplitude carries P;
  - its mirror $R \to -R$, since each frequency couples to its mirror through the packets;
  - in sweep and chain data, a constant channel with the time since the last pulse.
- **Output:** $U = (\sum U)\,\mathrm{softmax}(\ell)$, applied for T.
- **Loss:** KL divergence on the profile, plus 0.5× MSE on the standardized total and $\log T$.

With `-residual`, $\ell = \log \bar{\pi} + \ell_\theta$, where $\bar{\pi}$ is the mean training profile and the last layer is zero-initialized. The untrained network is then exactly the mean program, and training learns only the state-dependent correction.

Physics evaluation. Each prediction is first projected onto the caps: rescale $\sum U$ to the power cap, clip T, and shorten T to the energy cap. The projected program is then simulated from the stored populations. The score is the gain fraction $(Q_{\mathrm{net}} - Q_{\mathrm{noRF}})/(Q_{\mathrm{opt}} - Q_{\mathrm{noRF}})$, compared with two baselines:

- the mean training program;
- a nearest-neighbour lookup of the training spectrum.

Training data	Test	Network	Nearest neighb.	Mean program
Sweep, 240	held-out P_0 29, 33% (plain)	0.892	0.999	0.991
Sweep, 240	same, `-residual`	0.997	0.999	0.991
Chain, 70×6	14 new chains, residual	**0.991** (worst 0.90)	0.998	0.926 (worst 0.53)
Chain, 70×6	same, plain	0.978 (worst 0.52)	0.998	0.926

On closed-loop states the network learns state-specific pulses. Nearest-neighbour lookup is still slightly better at this dataset size, and it is the benchmark to beat as datasets grow.

21.2 Polarization model

`SpectrumToPQ` uses the same trunk. A pooled head outputs (P, Q), and a per-bin auxiliary head outputs I^+, with $I^- = I - I^+$. Supervising the branch split encodes $Q = A_+ - A_-$ (equation (28)). The baselines are the area method for P and $Q_{\mathrm{B}}(P)$ of equation (21). The Boltzmann relation is exact for unmanipulated states, so a useful network must beat it on manipulated ones.

22 Settings used in the present simulations

This section records the values behind every result in this note, so that each can be reproduced exactly.

22.1 Full parameter set of `d_butanol_matched`

Table 7: Every kinetic parameter of the default material.

Parameter	Value	Note
`n_bins, r_min, r_max`	500, ∓6.18620	$\Delta R = 0.024794$
`line_model`	multisite_butanol	
`line_gamma`	0.0235706	γ
`line_cd_scale_R`	1.0	C–D unit of R
`line_od_scale_ratio`	1.354745	ρ_{OD}
`line_eta_cd, line_eta_od`	0.0647698, 0.0326686	
`line_od_fraction`	0.0212736	f_{OD}
`line_phi_points`	128	azimuthal quadrature
`p0, q0`	0.50, Boltzmann	initial state
`rf_response_mode`	spectral	
`rf_spectral_strength`	0.0195741	κ [$\mathrm{R\,s^{-1}U^{-1}}$]
`rf_homogeneous_hwhm_R`	0.015	Γ
`rf_peak_selectivity`	0.7	s
`rf_peak_center_R`	0.935230	R_c
`rf_peak_half_width_R, rf_transition_width_R`	0.16, 0.25	selectivity band
`rf_coupling_*, rf_width_*, exchange_*, dnp_*, t1_* profiles`	neutral (exponents 0, $a_{2,4} = 0$, O–D factor 1, clip 12)	$f^X \equiv 1$
`diffusion_enabled`	true	
`diffusion_scale`	0.0128319	K_0 [$\mathrm{s^{-1}}$]
`zq_width_R, diffusion_overlap`	0.716884, lorentzian	w, S
`cross_branch_ratio`	1.0	r_{x}
`double_quantum_ratio`	3.032153	r_{DQ}
`orientation_corr_fraction, _width_deg`	0, 20	off
`kernel_cutoff_widths`	0	full tails
`microwave_diffusion_factor`	1.0	
`p_dnp_sat, dnp_rate`	0.475633, 0.00229831	P_{sat}, k_{D} [$\mathrm{s^{-1}}$]
`capacity_rate_power, _clip`	1.0, 12	DNP weighting
`t1_rate, t1_p_eq`	0, 0	off
`site_resolved, dipolar_enabled`	false, false	basic engine
`afp_efficiency`	1.0	
`dt`	0.03	s

The other materials change only the following:

- `d_butanol_basic`: `rf_peak_selectivity` 1.0, `rf_coupling_density_exponent` 0.5, `rf_width_theta_p2` 0.6.

- `d_butanol_extend`: selectivity 1.0, coupling exponent 1.0, `rf_width_density_exponent` −0.3, `rf_width_theta_p2` 0.6.
- `d_butanol_full_physics`: `site_resolved` and `dipolar_enabled` true, `dipolar_capacity_ratio` 0.01 ($\xi = 1$, $r_{1d} = 0$, $\beta_L = \beta_0 = 0$).
- `d_butanol_same_branch_transport`: $\kappa = 0.0108192$, $K_0 = 0.515827$, $w = 0.0471$, $r_{\mathrm{x}} = r_{\mathrm{DQ}} = 0$, $P_{\mathrm{sat}} = 0.450859$, $k_{\mathrm{D}} = 0.00227818$; monitor at $R = 1.9$.
- `demo_single_pake`: single site on $[-3, 3]$, $\Gamma_{\mathrm{line}} = 0.05$, $s = 0.04$, $\kappa = 0.04$, $R_c = 0.96$, $K_0 = 5$, $w = 0.05$, $r_{\mathrm{DQ}} = 0.1$, $P_{\mathrm{sat}} = 0.58$, $k_{\mathrm{D}} = 0.05$; arbitrary units.

22.2 Optimizer settings

Table 8: Optimizer settings: material default (GUI designer) and the `fast` preset with the pilot configuration (data generation).

`Setting`	Material default	Data generation	Meaning
`max_power`	20	20	per-bin ceiling $\bar{U}$
`max_total_power`	none	3 (0.3 W)	P_{cap}
`max_total_exposure`	none	102 (10.2 J)	E_{cap}
`min_duration`, `max_duration`	0.01, 10 s	0.5, 60 s	duration range
`duration_samples`	9	5	geometric scan
`duration_refinements`	3	1	bisections
`max_iterations`	100	40	per start
`starts`	2	1	seeds
`search_dt`	0.004	0.03	search step ceiling
`max_steps`	20000	20000	per horizon
`max_validation_halvings`	12	12	playback refinement
`max_wall_seconds`	300	180	time budget
`q_absolute / relative_tolerance`	10^{-6}, 0.002	same	selection tolerance
`convergence_tolerance`	2×10^{-5}	same	time-step validation
`candidate_mode`	union	union	candidate rule
`profile_smoothing_fwhm_R`	0.03	0	smoothing
`seed`	42	per sample	

The `standard` and `full` presets keep the material settings, with wall budgets of 600 s and 1800 s; `test` is for unit tests only.

Study conventions for the key results. Single-burn results start from a Boltzmann state at $P_0 = 0.50$ ($Q_0 = 0.1972$) with DNP off. They use two RF limits:

- *uncapped*: $U_j \leq 20$, with no total power or exposure limit;
- *capped*: $\sum U \leq 1$ (0.1 W) and $T \sum U \leq 34$ (3.4 J).

Continuous-DNP results set the DNP target to $P = 0.50$, so DNP alone holds $Q = 0.1972$.

22.3 Dataset generation runs

All three runs used `d_butanol_matched`, `-dnp on` (DNP during history, waits, optimized pulse and post-pulse evolution), `-power-cap 3`, `-exposure-cap 102`, `-preset fast` and `-config data/pilot_config.json`:

```
{"design": {"overrides": {"min_duration": 0.5, "max_duration": 60.0,
                          "search_dt": 0.03, "max_steps": 20000}},
 "history": {"wait": [0.0, 60.0]}}
```

Run	Settings	Size and cost
Sweep pilot (`sweep_dnp_on`)	`-p0-values 25 27 29 31 33 35`, `-histories-per-p0 8`, `-times 0 2 5 10 20`; one burn per history (centre $\|R\| \in [0, 2.2]$, width 0.02–0.6, total power 0.01–5 U, duration 0.01–20 s, all log-uniform); post-pulse 30 s in 3 steps with DNP; seed 12345	240 samples, all completed; 8714 s on 7 workers
Chain probe (`chain_probe`)	`-chains 14 -rounds 4 -p0-range 25 35`; history 0–1 burns (no wait) and, with probability 0.2, one AFP window (width 0.05–0.6, efficiency 0.7–1); no post-pulse recording; round wait 2–40 s log-uniform; undershoot power 0.3–1, duration 0.2–1, profile noise 0.5, exact 15%; seed 12345	56 samples, all completed; 1015 s
Chain dataset (`chain_v1`)	as the probe, `-chains 70 -rounds 6`, seed 2026	420 samples, all completed; 7262 s (≈208/h); mean design 117 s

22.4 Training runs

All runs used `python3 -m ml.train`: U-Net width 32, batch 32, AdamW (weight decay 10^{-4}) with cosine decay, loss weight 0.5 on the scalars, the best validation checkpoint, seed 0, the Apple GPU (`mps`). The physics evaluation integrates at $\min(0.03\,\text{s}, 0.2/\lambda_{\max})$.

Run	Options	Split
`ml_runs/sweep`	`-data sweep_dnp_on -epochs 300` (lr 2×10^{-3})	by group 80/10/10
`ml_runs/sweep_holdout`	`-holdout-p0 29 33 -lr 5e-4 -epochs 300`	test = P_0 29, 33%
`ml_runs/sweep_holdout_residual`	as above, `-residual`	same
`ml_runs/chain_v1_residual`	`-data chain_v1 -test-data chain_probe -epochs 400 -lr 5e-4 -residual`	train/val 372/48; test 56
`ml_runs/chain_v1_plain`	as above without `-residual`	same

22.5 Peak–pedestal evaluation

Table 4 was computed in the following setting:

- the full model's derivative, with DNP and RF off;
- a Boltzmann state at $P = 0.45$;
- a relative perturbation 10^{-4} of the a-mode in the packet nearest $R = 0.935$ (peak) or $R = -1.6$ (pedestal);
- $\lambda_{\mathrm{RF}}/U = 2W_k^+$ for a unit command on that bin.

23 Exact physics, phenomenological assumptions, and limitations

23.1 Relations that are exact within the spin-1 packet model

- The definitions of P, Q and the three populations, and the transition intensities as population differences.
- The mirror relation between the two spectral appearances of a packet.
- Conservation of packet spin count by every term.
- The equality of the two single-quantum RF matrix elements, and the one-half mirror rule for an isolated narrow carrier.
- $\dot{P}|_{\mathrm{RF}} = -\sum_k (J_k^+ + J_k^-)$ and $\dot{Q}|_{\mathrm{RF}} = 3\sum_k (J_k^- - J_k^+)$.
- The channel structure of spin-1 flip-flops, their equal matrix elements, and their conservation laws.
- The H-theorem and the uniqueness of the Boltzmann stationary state at the remaining P.
- Energy conservation of exchange with the dipolar reservoir.
- The discrete adjoint gradient of the optimizer's forward map.

23.2 Phenomenological choices

- The ZQ overlap shape and width, and the effective DQ ratio (a second-order process folded into one constant).
- Spatial mean-field pairing: the frequency-space kernel carries no real-space structure unless orientation correlation is enabled.
- The per-spin RF coupling and width profiles; the peak-selectivity band.
- DNP as relaxation to a Boltzmann target, with an empirical density weighting.
- The first-order (in $\beta\delta$) dipolar reservoir.
- Instantaneous AFP.
- One effective T_1, with an optional profile.

23.3 Important missing physics and extensions

1. **Coherent RF.** Nutation, spin locking and finite-rate passage, for $\omega_1 \gtrsim \Gamma$.
2. **Real-space transport.** Diffusion barriers around paramagnetic centres; spatially inhomogeneous DNP and microwave field.
3. **Explicit DNP channels.** Solid effect, cross effect and thermal mixing as separate electron–nuclear rate terms, and their combination when several act at once.
4. **Exponential reservoir coupling** for large $\beta\delta$.

5. **Complex CW-NMR response.** Absorption, dispersion, Q-meter phase, sweep timing and acquisition latency, for direct comparison with raw data.
6. **Global or time-shaped optimization.** The present designer is a local search over synchronized programs.
7. **The persistent ∼20% RF-on plateau at the peak** seen in data is not reproduced by any present parameter set.

New mechanisms enter at three places:

1. a term in `Spin1Model.derivative` (or `ExtendedCore.rhs`);
2. the same term, regrouped, in `FrozenDynamics.rhs`, together with its VJP;
3. its contribution to the outgoing-rate bound.

The existing tests are the template: compare the evaluators, check the adjoint against finite differences, and check the conservation laws.

24 Changes since the first release

Aspect	First release	Production release
RF drive	one ideal bin at one burn location; rate $\Gamma_{\mathrm{RF}} w_k$	arbitrary per-bin, time-scheduled programs; Lorentzian homogeneous kernel; per-spin coupling and width profiles
Density weighting	$w_k \propto \rho_k^{\alpha}$ on RF, DNP and recovery, on top of μ_k	capacity μ_k only, for RF and exchange; explicit mechanism profiles; density weight kept for DNP
Recovery	linear relaxation of two RF modes toward $\boldsymbol{n}^{\mathrm{B}}[P(t)]$, plus nearest-neighbour Gaussian diffusion	mass-action pair currents for same-branch, cross-branch and DQ channels on a nonlocal Lorentzian kernel
Conservation of P	enforced by projecting each internal term	by construction, pair by pair
Approach to Boltzmann	imposed by the reference state	derived: the unique stationary state
Sites, reservoir	none	optional C–D/O–D populations and dipolar reservoir
AFP	none	instantaneous swap with efficiency
Integration	fixed step; silent positivity clip	rate-bounded substeps, event splitting, error on real negatives
Design	none	adjoint-based, cap-constrained tensor optimizer with replay validation
Data / ML	none	command-line generator (random, sweep, chain) and PyTorch models
Calibration	dimensionless	d-butanol in seconds and watts
Grid	$N = 701$ on $[-3, 3]$	$N = 500$, symmetric, on $[-6.19, 6.19]$ for d-butanol

Tests	13	460

Two of these changes alter the physics, not only the numerics.

First, density is no longer counted twice. In the first release, w_k multiplied rates that already scale with μ_k through the populations, so absolute horn currents scaled roughly as ρ^2. The same double counting, in the RF law of the MaterialTuning-V1 development version, accounts for most of the difference between that version and the current per-spin law: $Q \approx 0.264$ versus 0.245–0.248 from $P_0 = 0.5$. The density law is still available, but as an explicit hypothesis about per-spin coupling.

Second, the first-release recovery relaxed toward a Boltzmann reference at the current P, which pulled Q toward $Q_{\mathrm{B}}(P)$ by construction. Its separate-mode neighbour flux also conserved Q exactly. In the production model, tensor order at fixed P relaxes only through the cross-branch channel, so its rate is a material parameter that can be measured. Whether real transport is same-branch or cross-branch is the largest open question for steady operation (section 11) and for the peak–pedestal contrast (section 18.7).

25 Conclusions

The ss-RF package models a spin-1 Pake doublet at the level of packet populations. Two structural features carry over from the first release:

- the two direct components at one frequency are different orientation packets;
- every direct hole is tied to its mirror response through shared populations.

The production release replaces the phenomenological recovery with the three spin-1 flip-flop channels, written as reversible pair currents. As a result, conservation of vector polarization, the distinction between spectral recovery and repolarization, and the approach to the Boltzmann line at the remaining P are consequences of the model rather than constraints imposed on it. The capacity of a packet and the rate per spin are kept separate: every per-spin dependence on frequency, orientation or site is an explicit, separately tunable profile.

On this model the software builds:

- a validated, adjoint-based designer of Q-maximizing RF programs under realistic power and energy limits;
- a closed-loop data generator that reproduces the operating cycle;
- learned models that reach 99% of the optimizer's gain on independent closed-loop states.

The calibrated d-butanol materials fit the lineshape, DNP build-up and one burn/refill experiment. Two quantities that matter most for tensor enhancement remain unconstrained: the per-spin RF selectivity and the locality of spin-diffusion transport. The peak–pedestal analysis shows that one experiment separates them. It should be done next: the required RF power at a matched fractional hole depth at peak and pedestal, together with the initial refill rates and a refill measurement in the I^--only region.

A Population-level differential rate equations and observable intensity evolution

This appendix collects the equations used by the simulation software and derives the quantities displayed in the graphical front end. It begins with the three spin-1 level populations in each orientation packet. It keeps the two packets that overlap at one physical frequency separate, and ends with the time derivatives of the plotted absorption components $I^+(R,t)$ and $I^-(R,t)$. The RF bookkeeping and the direct-hole/mirror-peak relation follow Ref. [23]. The recovery equations implement the spectral-overlap transport of Ref. [24] as pair currents.

A.1 The state variable: three populations in each spin packet

A spin-1 nucleus has three Zeeman sublevels $m = +1, 0, -1$. The simulation divides the broadened line into N packets indexed by k, at $x_k = R_+(\theta_k)$, each with the population vector

$$\boldsymbol{n}_k(t) = \begin{pmatrix} n_{k+}(t) \\ n_{k0}(t) \\ n_{k-}(t) \end{pmatrix}, \qquad n_{k+} + n_{k0} + n_{k-} = \mu_k, \qquad \sum_{k=0}^{N-1} \mu_k = 1. \tag{103}$$

The fractional populations within a packet are $p_{km} = n_{km}/\mu_k$. The n_{km} are absolute population fractions of the whole sample; the p_{km} describe how the spins inside packet k are divided among the levels. With the site-resolved engine, the state is $n_{s,k,m}$ with $\sum_s n_{s,k,m} = n_{km}$.

A.2 Why one physical frequency contains two different population vectors

Packet k appears at $R = x_k$ in I^+ and at $R = -x_k$ in I^-. A frequency R_b in the overlap region therefore contains two generally different packets:

$$A \equiv k_+(R_b) : \; x_A \simeq R_b, \qquad B \equiv k_-(R_b) = \bar{A} : \; x_B \simeq -R_b. \tag{104}$$

With $\mathcal{K} \equiv \mathcal{C}_{\text{disp}}/\Delta R$, the four direct and mirror observables associated with the pair are

$$I^+(R_b, t) = \mathcal{K}\,[n_{A+}(t) - n_{A0}(t)], \tag{105}$$
$$I^-(R_b, t) = \mathcal{K}\,[n_{B0}(t) - n_{B-}(t)], \tag{106}$$
$$I^-(-R_b, t) = \mathcal{K}\,[n_{A0}(t) - n_{A-}(t)], \tag{107}$$
$$I^+(-R_b, t) = \mathcal{K}\,[n_{B+}(t) - n_{B0}(t)]. \tag{108}$$

The signal at $-R_b$ is not a second, independent reservoir. It is another measurement of a population difference in the same packet that was modified at $+R_b$. At $R_b = 0$, $A = B$, and one packet contributes both transitions at the same frequency.

A.3 Elementary two-level equalization

For two levels with a symmetric saturation process,

$$J_{ab} = W_{ab}(n_a - n_b), \qquad \dot{n}_a = -J_{ab}, \qquad \dot{n}_b = +J_{ab}, \qquad \frac{\mathrm{d}}{\mathrm{d}t}(n_a - n_b) = -2W_{ab}(n_a - n_b). \tag{109}$$

The measured transition difference therefore decays at $2W_{ab}$, twice the coefficient in the symmetric current. The sign is automatic: if $n_a < n_b$, population flows the other way.

A.4 The RF currents with a finite homogeneous linewidth

For a program with carrier commands $U_j(t)$ and the RF gate $u_{\text{RF}} \in \{0, 1\}$,

$$J_k^+ = u_{\text{RF}}\, c_k \Big[\sum_j K_k(x_k - x_j) U_j\Big](n_{k+} - n_{k0}), \tag{110}$$

$$J_k^- = u_{\text{RF}}\, c_k \Big[\sum_j K_k(-x_k - x_j) U_j\Big](n_{k0} - n_{k-}), \tag{111}$$

with $K_k(\delta) = \kappa\Gamma_k/(\Gamma_k^2 + \delta^2)$. The superscript on J_k^+ denotes the $+ \leftrightarrow 0$ transition and on J_k^- the $0 \leftrightarrow -$ transition. In the ideal-bin limit with one carrier at R_b, only J_A^+ and J_B^- are nonzero. In general, every packet within a few Γ of any carrier, or of its mirror, is driven.

A.5 The internal (exchange) currents

With $\tilde{S}_{ij} = \sqrt{f_i^{\mathrm{ex}} f_j^{\mathrm{ex}}}\, S(x_i - x_j)$, $\tilde{X}_{ij} = \sqrt{f_i^{\mathrm{ex}} f_j^{\mathrm{ex}}}\, S(x_i + x_j)(1 - \frac{1}{2}\delta_{ij})$, and $\tilde{C}_{ij} = \sqrt{f_i^{\mathrm{ex}} f_j^{\mathrm{ex}}}\, C_{ij}$:

$$G_{k+}^{\mathrm{SQ}} = K_0\big[n_{k0}(\tilde{S}n_+)_k - n_{k+}(\tilde{S}n_0)_k\big], \tag{112}$$
$$G_{k-}^{\mathrm{SQ}} = -K_0\big[n_{k-}(\tilde{S}n_0)_k - n_{k0}(\tilde{S}n_-)_k\big], \tag{113}$$
$$G_{k0}^{\mathrm{SQ}} = -G_{k+}^{\mathrm{SQ}} - G_{k-}^{\mathrm{SQ}}, \tag{114}$$
$$G_{k+}^{\mathrm{X}} = r_{\mathrm{x}} K_0\big[n_{k0}(\tilde{X}n_0)_k - n_{k+}(\tilde{X}n_-)_k\big], \tag{115}$$
$$G_{k-}^{\mathrm{X}} = r_{\mathrm{x}} K_0\big[n_{k0}(\tilde{X}n_0)_k - n_{k-}(\tilde{X}n_+)_k\big], \tag{116}$$
$$G_{k0}^{\mathrm{X}} = -G_{k+}^{\mathrm{X}} - G_{k-}^{\mathrm{X}}, \tag{117}$$
$$G_{k+}^{\mathrm{DQ}} = -G_{k-}^{\mathrm{DQ}} = r_{\mathrm{DQ}} K_0\big[n_{k-}(\tilde{C}n_+)_k - n_{k+}(\tilde{C}n_-)_k\big], \qquad G_{k0}^{\mathrm{DQ}} = 0. \tag{118}$$

Each line follows from summing the pair currents of equations (67) to (69) and (71) over partners j. With DNP on, K_0 carries the microwave factor. Summed over k:

- $\sum_k G_{km}^{\mathrm{SQ}} = 0$ for each m;
- $\sum_k (G_{k+}^{\mathrm{X}} - G_{k-}^{\mathrm{X}}) = 0$, using the symmetry of $\tilde{X}$;
- $\sum_k G_{k\pm}^{\mathrm{DQ}} = 0$.

These are the conservation laws of section 7.4, with no projection.

A.6 Full population equations

Collect all non-RF terms into

$$G_{km} = G_{km}^{\mathrm{SQ}} + G_{km}^{\mathrm{X}} + G_{km}^{\mathrm{DQ}} + G_{km}^{\mathrm{DNP}} + G_{km}^{T_1}, \tag{119}$$

with

$$G_{km}^{\mathrm{DNP}} = u_{\mathrm{DNP}}\, k_{\mathrm{D}} w_k^{\mathrm{D}} \left(\mu_k p_m^{\mathrm{B}}(P_{\mathrm{sat}}) - n_{km}\right), \tag{120}$$
$$G_{km}^{T_1} = k_1 f_k^{T_1} \left(\mu_k p_m^{\mathrm{B}}(P_{\mathrm{eq}}) - n_{km}\right). \tag{121}$$

The complete population equations are

$$\dot{n}_{k+} = -J_k^+ + G_{k+}, \tag{122}$$
$$\dot{n}_{k0} = +J_k^+ - J_k^- + G_{k0}, \tag{123}$$
$$\dot{n}_{k-} = +J_k^- + G_{k-}. \tag{124}$$

The RF part conserves the packet spin count, because the three RF contributions sum to zero.

In the ideal-bin limit at $R_b \neq 0$, the directly addressed packets satisfy

$$\dot{n}_{A+} = -J_A^+ + G_{A+}, \qquad \dot{n}_{B+} = G_{B+}, \tag{125}$$
$$\dot{n}_{A0} = +J_A^+ + G_{A0}, \qquad \dot{n}_{B0} = -J_B^- + G_{B0}, \tag{126}$$
$$\dot{n}_{A-} = G_{A-}, \qquad \dot{n}_{B-} = +J_B^- + G_{B-}. \tag{127}$$

This is the simplest complete answer to the statement that "the populations at R and $-R$ are different." With a finite linewidth, the same structure holds for every driven packet.

A.7 Vector and tensor polarization at the population level

$$P(t) = \sum_k [n_{k+} - n_{k-}], \qquad Q(t) = \sum_k [n_{k+} - 2n_{k0} + n_{k-}], \tag{128}$$

$$\dot{P} = -\sum_k (J_k^+ + J_k^-) + \sum_k (G_{k+}^{\mathrm{DNP}} - G_{k-}^{\mathrm{DNP}} + G_{k+}^{T_1} - G_{k-}^{T_1}), \tag{129}$$

$$\dot{Q} = 3\sum_k (J_k^- - J_k^+) - 3\sum_k G_{k0}^{\mathrm{X}} + \sum_k (G_{k+}^{\mathrm{DNP}} - 2G_{k0}^{\mathrm{DNP}} + G_{k-}^{\mathrm{DNP}}) + (T_1 \text{ analog}). \tag{130}$$

Exchange does not appear in $\dot{P}$. Only the cross-branch channel appears in $\dot{Q}$, where $-3\sum_k G_{k0}^{\mathrm{X}} = 6\sum_{k,j} J_{kj}^{(\mathrm{x})}$, which vanishes when $p_0^2 = p_+ p_-$.

A.8 Linearized local refill

For a small I^+ hole in packet k ($p_{k+} = \bar{p}_+ - \delta$, $p_{k0} = \bar{p}_0 + \delta$), with all partners at the Boltzmann fractions $\bar{p}$, equations (112) and (113) give

$$\dot{p}_{k+}\big|_{\mathrm{SQ}} = K_0[(\tilde{S}\mu)_k - \tilde{S}_{kk}\mu_k](\bar{p}_+ + \bar{p}_0)\,\delta, \tag{131}$$

$$\dot{p}_{k0}\big|_{\mathrm{SQ}} = -K_0[(\tilde{S}\mu)_k - \tilde{S}_{kk}\mu_k](\bar{p}_+ + \bar{p}_0 + \bar{p}_-)\,\delta, \tag{132}$$

so that $d_{k,+} = p_{k+} - p_{k0}$ relaxes at $\kappa_k^{\mathrm{SQ}} = K_0[(\tilde{S}\mu)_k - \tilde{S}_{kk}\mu_k](1 - \bar{p}_-/2)$, which is equation (102). The cross-branch and DQ contributions follow the same way, with $\tilde{X}$ and $\tilde{C}$.

A.9 From population equations to the plotted intensity rates

For any packet,

$$\boxed{\dot{I^+}(x_k) = \mathcal{K}\left[-2J_k^+ + J_k^- + G_{k+} - G_{k0}\right]} \tag{133}$$

$$\boxed{\dot{I^-}(-x_k) = \mathcal{K}\left[+J_k^+ - 2J_k^- + G_{k0} - G_{k-}\right].} \tag{134}$$

At the monitor marker $M = R_b$ (packets A, B), the solid traces obey

$$\boxed{\frac{\mathrm{d}I^+(R_b,t)}{\mathrm{d}t} = \mathcal{K}\left[-2J_A^+ + J_A^- + G_{A+} - G_{A0}\right]} \tag{135}$$

$$\boxed{\frac{\mathrm{d}I^-(R_b,t)}{\mathrm{d}t} = \mathcal{K}\left[J_B^+ - 2J_B^- + G_{B0} - G_{B-}\right]} \tag{136}$$

and the dotted mirror traces obey

$$\boxed{\frac{\mathrm{d}I^-(-R_b,t)}{\mathrm{d}t} = \mathcal{K}\left[J_A^+ - 2J_A^- + G_{A0} - G_{A-}\right]} \tag{137}$$

$$\boxed{\frac{\mathrm{d}I^+(-R_b,t)}{\mathrm{d}t} = \mathcal{K}\left[-2J_B^+ + J_B^- + G_{B+} - G_{B0}\right].} \tag{138}$$

In the ideal-bin limit $J_A^- = J_B^+ = 0$, and the first-release forms are recovered.

A.10 RF-only limit and the one-half mirror-gain rule

With all G set to zero and the ideal-bin limit,

$$\dot{I^+}(R_b)\big|_{\mathrm{RF}} = -2\mathcal{K}J_A^+, \qquad \dot{I^-}(-R_b)\big|_{\mathrm{RF}} = +\mathcal{K}J_A^+, \tag{139}$$

$$\dot{I^-}(R_b)\big|_{\mathrm{RF}} = -2\mathcal{K}J_B^-, \qquad \dot{I^+}(-R_b)\big|_{\mathrm{RF}} = +\mathcal{K}J_B^-, \tag{140}$$

so the opposing-branch area gained is one half of the directly burned area lost [23]. With a finite linewidth, J_A^- and J_B^+ are nonzero near $R = 0$, and the wings drive neighbouring packets. The exact relations are then equations (135) to (138).

A.11 Mechanism-by-mechanism intensity decomposition

`derivative(breakdown=True)` returns, for the monitored packets, the contribution of each term to $\dot{I^+}$, $\dot{I^-}$ (direct and mirror), $\dot{P}$ and $\dot{Q}$:

$$\dot{I^+}_k = \dot{I^+}_k^{\mathrm{RF}} + \dot{I^+}_k^{\mathrm{SQ+}} + \dot{I^+}_k^{\mathrm{SQ-}} + \dot{I^+}_k^{\mathrm{X}} + \dot{I^+}_k^{\mathrm{DQ}} + \dot{I^+}_k^{\mathrm{DNP}} + \dot{I^+}_k^{T_1}, \tag{141}$$

with, for each term Y, $\dot{I^+}_k^{\,Y} = \mathcal{K}(G_{k+}^Y - G_{k0}^Y)$ and $\dot{I^-}_k^{\,Y} = \mathcal{K}(G_{k0}^Y - G_{k-}^Y)$. These are the natural quantities to compare with measured short-time slopes. They are also the tool for separating RF response from replenishment in the peak–pedestal tests.

A.12 Numerical time step and constraints

The equations are advanced with the adaptive, event-split Euler step of equation (88). A population below $-10^{-13}\mu_k$ is an error. Tiny rounding negatives are floored at 10^{-30}, and each packet is rescaled to μ_k:

$$\boldsymbol{n}_k \leftarrow \mu_k \frac{\boldsymbol{n}_k}{\sum_m n_{km}}. \tag{142}$$

These operations are numerical safeguards, not physical source terms. The step bound $h \le 0.2/\lambda_{\max}$ guarantees positivity for the linear terms, independently of the command amplitude, so commands are never silently corrected by the clamp.

A.13 Operational summary

At every substep, the software performs the following sequence:

1. determine the next pulse edge and the active command vector U;
2. compute $W_k^{\pm}$ from U by convolution, and the RF currents $J_k^{\pm}$ from the current population differences;
3. compute the exchange fields Sn, Xn and Cn by FFT and $O(N)$ sums, and the currents G^{SQ}, G^{X}, G^{DQ};
4. add the DNP and T_1 terms (and, in the extended engine, the reservoir terms and $\dot{\beta}$);
5. choose h from the rate bound and the next edge, advance, check positivity, renormalize, and accumulate exposure;
6. project the populations to $I^+(R,t)$ and $I^-(R,t)$ through equations (105) to (108) and their full-grid equivalents.

B Interpretation of the main rate controls

Parameter	Symbol	Recommended interpretation
`rf_spectral_strength`	κ	RF rate scale per unit command, in R units; κ/Γ is the on-resonance rate per U.
`rf_homogeneous_hwhm_R`	Γ	Homogeneous (T_2) half-width of a packet's transitions; sets hole width.
`rf_peak_selectivity`	s	Density exponent of per-spin RF coupling inside the peak bands; not fitted.
`rf_peak_center_R`, `rf_peak_half_width_R`, `rf_transition_width_R`	R_c	Location and taper of the peak bands.
`rf_coupling_pedestal_gain`	G_{p}	Per-spin coupling outside the peak bands.
`X_density_exponent`, `X_theta_p2/p4`, `X_od_factor`	e_X, $a^X_{2,4}$, m^X_{OD}	Mechanism profile, equation (57), for X in rf_coupling, rf_width, exchange, dnp, t1.
`diffusion_scale`	K_0	Flip-flop rate scale; refill rate is K_0 times the partner count.
`zq_width_R`, `diffusion_overlap`	w, S	ZQ overlap width and shape; narrow w makes transport local and density-dependent.
`cross_branch_ratio`	r_{x}	Cross-branch / same-branch; 1 from equal matrix elements; the only channel that relaxes Q at fixed P.
`double_quantum_ratio`	r_{DQ}	Effective second-order DQ exchange; nonlocal vector transport.
`microwave_diffusion_factor`	—	Multiplies K_0 while DNP is on.
`orientation_corr_fraction`, `_width_deg`	f_θ, σ_θ	Optional orientation–space correlation of exchange partners.
`kernel_cutoff_widths`	—	Optional hard cutoff of the overlap in units of w; 0 keeps all tails.
`dnp_rate`, `p_dnp_sat`	$k_{\mathrm{D}}, P_{\mathrm{sat}}$	DNP build rate and target.
`capacity_rate_power`, `_clip`	α_{D}, c	DNP density weighting (legacy RF modes also use it).
`t1_rate`, `t1_p_eq`	k_1, P_{eq}	Spin-lattice relaxation and its target.
`site_resolved`, `exchange_intersite_factor`	M_{12}	Separate C–D/O–D populations and their mutual exchange.
`dipolar_enabled`, `dipolar_capacity_ratio`	c_d	Dipolar reservoir and its heat capacity.
`dipolar_exchange_coupling`	ξ	Flip-flop coupling to the reservoir.
`dipolar_relaxation_rate`, `dipolar_beta_lattice`	r_{1d}, β_L	Reservoir relaxation.
`afp_efficiency`	ε	AFP swap efficiency.
`dt`	δt	Material step; substeps are chosen automatically.

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